# Three-Dimensional Finite-Element Modeling of Microwave Ablation of a Patient-Derived Liver Tumor Using a Dual-Slot Coaxial Antenna

Electromagnetic Energy Deposition, Pennes Bioheat Transfer, Arrhenius Injury Modeling, Kinetic Sensitivity, and Fixed-Placement Power–Duration Analysis

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**Abstract**

A three-dimensional finite-element workflow was developed to investigate microwave heating and model-based thermal injury in an irregular, image-derived liver tumor surrounding a dual-slot coaxial applicator. The electromagnetic problem is solved at 2.45 GHz using first-order Nédélec edge elements in MFEM, distributed sparse direct factorization (MUMPS), and a TEM Robin feed. A conforming Gmsh mesh contains 294,380 tetrahedra, of which 226,153 belong to the electromagnetic solve. The algebraic relative residual is $3.17 \times 10^{-12}$; the projected port coefficient is $S_{11} \simeq -11.90\,\mathrm{dB}$, while the independently assessed direct-flux power-balance mismatch is approximately 0.2805%. For a nominal 10 W incident excitation, the power conservatively mapped into a discontinuous piecewise-constant heat-source field totals 9.004,97 W: 5.656,43 W in the tumor and 3.348,54 W in surrounding liver tissue. A separate parallel transient Pennes solver uses piecewise tissue density, specific heat and conductivity, a constant perfusion rate of $0.0036\,\mathrm{s}^{-1}$, insulated thermal boundaries and implicit Euler time stepping with $\Delta t = 1\,\mathrm{s}$. After 300 s, the predicted maximum tissue temperature reaches 85.728 °C; instantaneous perfusion cooling is approximately 5.865 W. A spatial-quadrature Arrhenius calculation predicts that 30.651% of the tumor volume reaches the chosen model-injury criterion $\theta_d \geq 0.98$ ($4.115\,\mathrm{cm}^3$), accompanied by $0.212{,}3\,\mathrm{cm}^3$ of healthy-tissue volume above the same criterion. Substitution of an alternative in-vitro HepG2 kinetics yields only 7.481% tumor coverage, while other independent and paired parameter perturbations give markedly different predictions. A subsequent fixed-placement power–duration sweep compared six completed numerical protocols (10–12 W; 300–600 s): predicted HCC-reference tumor injury-criterion coverage ranged from 30.651% to 52.700%, with corresponding healthy-tissue injury volumes from $0.212{,}3\,\mathrm{cm}^3$ to $1.448{,}3\,\mathrm{cm}^3$. At 12 W for 300 s, the model yielded 39.258% coverage and 95.47 °C peak temperature, highlighting the limitations of fixed-property high-temperature modeling. Electromagnetic accounting and thermal discrete energy balances have been checked, but spatial/time convergence and experimental validation remain outstanding. Injury fractions are *model-dependent surrogate outputs*, not histologically confirmed necrosis or a clinically validated treatment plan.

**Keywords:** microwave ablation; liver tumor; dual-slot coaxial antenna; H(curl) finite elements; Nédélec elements; Gmsh; MFEM; MUMPS; electromagnetic dosimetry; conservative source transfer; Pennes bioheat; Arrhenius thermal injury; sensitivity analysis.

# 1 Introduction

Microwave ablation (MWA) seeks to deposit electromagnetic energy within target tissue while minimizing injury to adjacent normal tissue. Applicator geometry, dielectric heterogeneity, impedance matching, and boundary placement jointly affect the spatial distribution of energy deposition. For a heterogeneous, patient-derived tumor geometry, rotational symmetry is generally unavailable, motivating a full three-dimensional model rather than the axisymmetric approximation used in many reference examples.

The dual-slot coaxial configuration investigated by Brace [1] provided the geometrical motivation for the present design: a proximal annular slot of 1 mm, a distal annular slot of 4 mm, and an 8 mm intervening section. The present work *adopts those nominal slot dimensions* but does not reproduce that study's experimental setup or claim equivalent thermal performance. A COMSOL reference model [4] was consulted for the general workflow from microwave fields to tissue heating; its reference geometry is a *single-slot, two-dimensional axisymmetric antenna*, not the three-dimensional dual-slot system studied here.

This report records both the final formulation and the engineering decisions needed to obtain a conforming mesh, a stable distributed solution, a trustworthy energy audit, and a power-conservative heat-source field. Particular attention is given to the distinction between nominal incident, modal, and net delivered powers. The manuscript now also includes completed transient thermal, Arrhenius-injury and kinetic sensitivity calculations from the same verified electromagnetic source; all results remain subject to the stated modeling limitations.

## 1.1 Scope and research questions

The computational study addresses seven questions: (i) can an irregular segmented tumor be incorporated with conforming interfaces; (ii) is the 3D vector Maxwell field sufficiently well resolved for meaningful power accounting; (iii) how sensitive are the feed and boundary observables to discretization; (iv) does electromagnetic source transfer preserve absorbed power; (v) how does that power produce temperature rise and model-based injury over time; (vi) how robust are the predicted ablation metrics to tissue-specific Arrhenius kinetics; and (vii) how do power and duration change injury-criterion coverage and modeled healthy-tissue damage when the applicator position is fixed?

# 2 Geometry and model development

## 2.1 Dual-slot antenna dimensions

The applicator is oriented along the Cartesian $x$-axis, with the coaxial feed at $x = 0$. The principal dimensions used in the current Gmsh/OpenCASCADE script are reported in Table 1. The feed consists of a central conductor, an inner coaxial dielectric, a tubular outer conductor interrupted by two annular slots, a metallic distal cap and conical tip, and a surrounding insulating catheter. The catheter is modeled as a distinct dielectric region, not as conducting tissue.

Table 1: Actual design inputs from `antena_V3_2_tumor107_R35_port_h010.geo`. All lengths are in millimetres.

| Quantity | Value | Quantity | Value |
|---|---|---|---|
| Inner conductor radius $a$ | 0.25 | Inner dielectric outer radius $b$ | 0.80 |
| Outer metallic conductor radius | 1.00 | Catheter outer radius | 1.20 |
| Straight coaxial length | 50.0 | Catheter thickness (radial) | 0.20 |
| Proximal slot width $G_1$ | 1.0 | Proximal slot interval | $x = 37$–$38$ |
| Slot separation $L$ | 8.0 | Distal slot interval | $x = 46$–$50$ |
| Distal slot width $G_2$ | 4.0 | Metallic cap length | 0.5 |
| Conical metallic tip length | 1.0 | Outer tissue cylinder length | 71.7 |
| Outer tissue radius (final) | 35.0 | Initial/earlier radii | 20 / 27 |

The slot positions follow the script definitions $x_{G_2} = 50 - G_2 = 46$ mm and $x_{G_1} = x_{G_2} - L - G_1 = 37$ mm. The distal metal reaches approximately $x = 51.5$ mm, and the computational tissue cylinder extends to $x = 71.7$ mm. The antenna and catheter dimensions remain unchanged between the R27 and R35 sensitivity runs; the latter modify the surrounding domain and mesh.

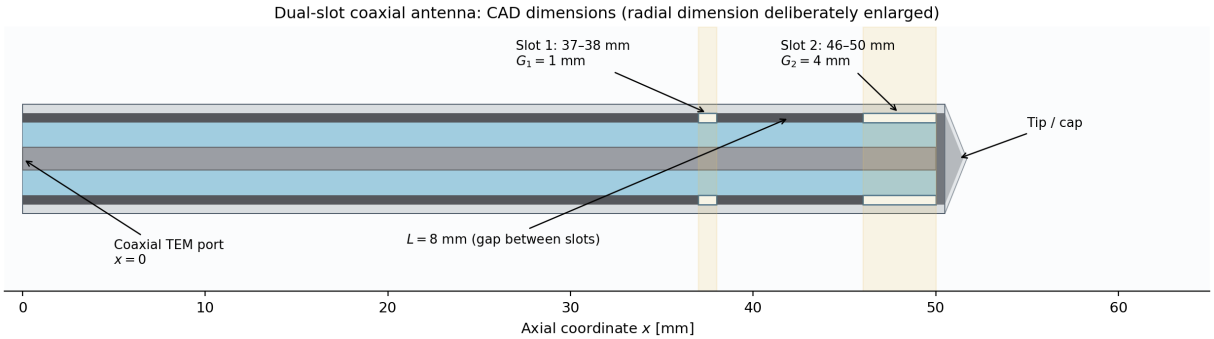



Figure 1: Schematic of the modeled dual-slot antenna, constructed from the parameters in the final Gmsh script. Radial dimensions are **deliberately exaggerated** for visibility; the drawing is not a patient-specific reconstruction.

## 2.2 Patient-derived tumor and conforming construction

The patient-derived tumor geometry was obtained from case `3Dircadb1.2` of the publicly available 3D-IRCADb-01 dataset [3]. The supplied case contains 172 abdominal CT slices (`PATIENT_DICOM.zip`), corresponding anatomical segmentation masks (`MASKS_DICOM.zip`), and reconstructed surface meshes. In particular, the tumor segmentation is available under the label `livertumor`, while its three-dimensional surface geometry is provided as `livertumor.vtk` and `tumor.stl`.

Inspection of the original surface mesh confirmed that it consists of 2,502 vertices and 5,000 triangular facets forming a single connected surface. Its original bounding box is approximately $x \in [113.56, 145.18]$ mm, $y \in [148.86, 182.79]$ mm, and $z \in [89.60, 124.80]$ mm, reflecting the coordinate system of the supplied dataset.

For the electromagnetic model, the tumor surface underwent remeshing with a target characteristic element size of approximately 1.5 mm, followed by BREP conversion and positioning relative to the coaxial antenna. The geometry was subsequently adapted to accommodate the catheter while preserving the conforming interface between malignant and healthy tissue. In the final computational configuration, the tumor extends approximately from $x = 27.74$ mm to $x = 59.26$ mm along the antenna axis. The resulting geometry is stored as `tumor_h1_5_107_izvan_katetera_m.brep` and assigned physical volume attribute 107.

Figure 2 shows the remeshed tumor used in the geometry-preparation workflow. The visible triangular facets describe its exterior; a surface rendering alone cannot establish whether a computational region contains volume elements. In the final simulation mesh, material 107 contains 51,667 tetrahedra, as documented in Table 7. The applicator channel is an intentional geometric exclusion, not removal of the tumor interior.

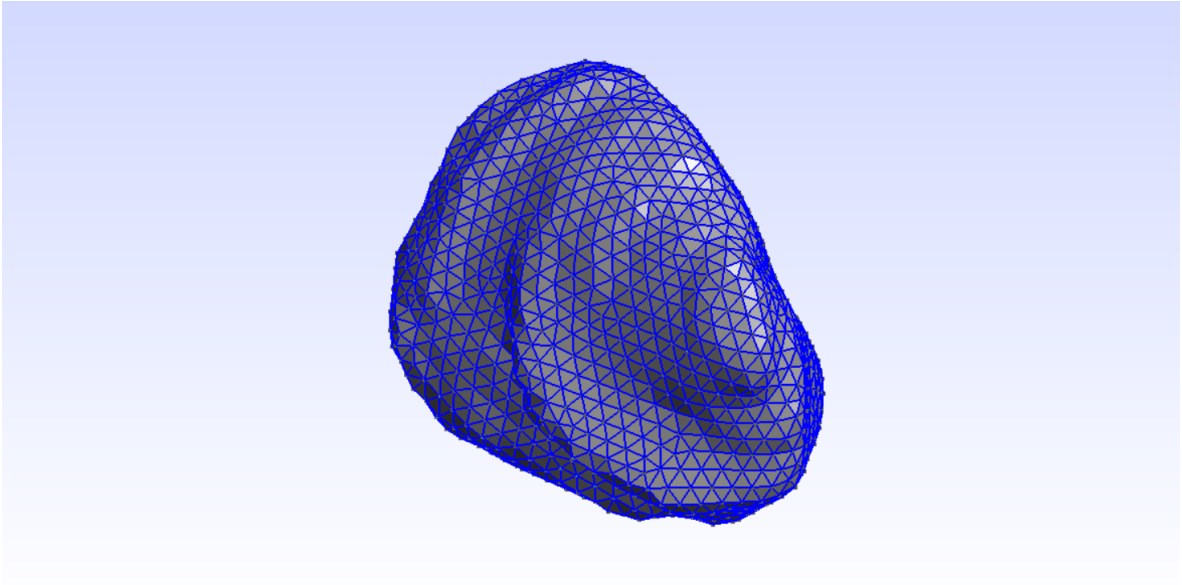


Figure 2: User-supplied view of the remeshed patient-derived tumor before incorporation into the antenna–tissue model. The exterior triangulation shows the irregular lesion shape used in the subsequent CAD construction; it is distinct from the finer conforming mesh of the final simulation.

Three CAD corrections were critical. First, the tumor BREP already stored positions in *metres*; the Gmsh line `Geometry.OCCScaling = 1` prevents an accidental additional $10^{-3}$ scaling. Second, a volume representing the catheter envelope was subtracted from the tumor before import, avoiding physical overlap between malignant tissue and the applicator. Third, the tumor and existing antenna/tissue solids were subjected to a `BooleanFragments` operation. The solids were subsequently selected again by geometry-dependent bounding boxes because OpenCASCADE Boolean operations can invalidate earlier entity identifiers. Material physical groups were only assigned after this final fragmentation. Gmsh documents `BooleanFragments` as creating conforming interfaces [6].

The final tumor occupies 13,425.216,396 $\mathrm{mm}^3$ and the final healthy surrogate cylinder contains 261,833.000,277 $\mathrm{mm}^3$ of tissue after excluding antenna, catheter, and tumor. These are *model-region volumes*: the enclosing cylinder does not constitute an anatomical segmentation of the whole liver. Tumor/healthy interfaces share 10,056 triangular faces, and tumor/catheter interfaces share 8,114 triangular faces in the final mesh. Thus, the interfaces are conforming, rather than formed by intersecting independent meshes.

### 2.3 Why the antenna and meshing script were revised

The development history is summarized in Table 2. It distinguishes *physical geometry changes* from *mesh, CAD bookkeeping, and numerical boundary changes*. This distinction is essential when attributing changes in absorbed power to any one cause.

Table 2: Traceable modifications documented in the available scripts and numerical logs; results are not from a controlled antenna-shape optimization study.

| Stage | Change | Technical motivation |
|---|---|---|
| Initial coaxial CAD | Inner/outer conductors, two slots, metal cap, conical tip, catheter; cylinder of tissue (initial radius 20 mm) | Construct physically separate dielectric/metallic domains and represent the dual-slot radiator. |
| V2 targeted mesh | Refined inner conductor to $h = 0.12$ mm, slots to $h = 0.15$ mm; coarse far field up to $h = 4$ mm | Resolve small conductor cross-section and narrow annular gaps without globally excessive elements. |
| V3 mesh optimization | Localized fine-mesh boxes; catheter-near target raised to $h = 0.55$ mm; shorter transition thickness | Reduce model size and MUMPS factorization cost while keeping resolution at electromagnetically important interfaces. |
| V3.1–V3.2 | Added thin MathEval annular refinement at $r = 0.80$, 1.00, and 1.20 mm, with respective targets 0.24, 0.25, 0.30 mm | Improve approximation of coaxial dielectric, outer-conductor, and catheter circumferences/PEC boundaries. |
| Tumor integration | Remeshed tumor BREP; catheter envelope subtracted; conforming `BooleanFragments`; material 107; recovered physical tags | Eliminate solid overlap, conform material interfaces, and provide distinct tumor dielectric properties. |
| BREP scale correction | `Geometry.OCCScaling = 1`; explicit tumor bounding-box checks | Prevent factor-of-1000 scaling errors inherited from earlier scripts. |
| R27 → R35 | Tissue outer radius enlarged from 27 to 35 mm, with original antenna and tumor geometry fixed | Test sensitivity to artificial absorbing boundary placement. |
| Fine-port R35 | Added `Field[11]`: $h_{\text{port}} = 0.10$ mm over $x \simeq -0.10$ to 0.80 mm; decreased `Mesh.MeshSizeMin` to 0.10 mm | Resolve the $x = 0$ coaxial annulus and reduce boundary/modal power discrepancies. |

## 2.4 Physical regions and boundary identifiers

The final Gmsh volume identifiers and numerical boundary identifiers are preserved throughout mesh extraction and assembly (Table 3). The metal is retained in the geometry for conforming meshing, then excluded from the electromagnetic *solve* using `SubMesh::CreateFromDomain`. Its physical effect is imposed through perfect electric conductor (PEC) boundaries. Boundary 204 is identified on the open catheter inlet when the metal-free submesh is constructed; it is not an independently listed physical surface in the original mesh.

Table 3: Material/port identifiers in the current CAD and C++ source.

| ID | Material/condition | Interpretation |
|---|---|---|
| 101 | Metal | Omitted from electromagnetic solution; creates PEC interfaces |
| 102 | Coaxial dielectric | Inner feed insulation |
| 103 | Slot 1 (air) | Proximal annular radiator |
| 104 | Slot 2 (air) | Distal annular radiator |
| 105 | Catheter dielectric | Applicator sheath |
| 106 | Healthy liver | Lossy tissue surrogate |
| 107 | Tumor | Lossy segmented tumor region |
| 201 | Incident TEM Robin | Coaxial dielectric feed at $x = 0$ |
| 202 | Perfect electric conductor | Tangential electric field vanishes on metal interfaces |
| 203 | Absorbing Robin | Outer healthy-tissue boundary |
| 204 | Absorbing Robin | Open catheter inlet, detected on EM submesh |

# 3 Electromagnetic formulation

## 3.1 Complex Maxwell curl–curl problem

With phasor convention $\exp(+i\omega t)$ and isotropic, piecewise constant material parameters, Maxwell's equations yield

$$\nabla \times (\mu^{-1}\nabla \times \mathbf{E}) - \omega^2 \varepsilon \mathbf{E} + i\omega\sigma \mathbf{E} = \mathbf{0}, \qquad \omega = 2\pi f, \quad f = 2.45\,\text{GHz}. \tag{1}$$

Equivalently, the complex permittivity is $\tilde{\varepsilon} = \varepsilon - i\sigma/\omega$. The numerical program uses separate real-valued piecewise coefficients for $\mu^{-1}$, $-\omega^2\varepsilon$, and $+\omega\sigma$, and assembles them with `CurlCurlIntegrator` and complex `VectorFEMassIntegrator` instances. Everywhere $\mu_r = 1$.

Table 4: Frequency-domain material constants *as implemented* at 2.45 GHz; no thermal dependence is included.

| ID | Material | Electromagnetic role | $\varepsilon_r$ | $\sigma$ [S/m] | $\mu_r$ |
|---|---|---|---|---|---|
| 102 | Coax dielectric | lossless | 2.03 | 0 | 1 |
| 103 | Proximal slot | air | 1.00 | 0 | 1 |
| 104 | Distal slot | air | 1.00 | 0 | 1 |
| 105 | Catheter | lossless | 2.60 | 0 | 1 |
| 106 | Healthy liver | lossy | 57.55 | 1.95 | 1 |
| 107 | Malignant liver region | lossy | 62.44 | 2.18 | 1 |

The code comments associate malignant-tissue values with the in-vivo liver measurements of O'Rourke *et al.* [2]. The precise provenance of the healthy-liver pair $(57.55, 1.95)$ should be explicitly documented against the original dataset or publication before journal submission. Tumor-specific permittivity and conductivity are *assumed constants*, not measurements made on the IRCAD patient's tumor. Tissue dielectric properties vary with pathology, temperature, and measurement conditions; a single pair per tissue is a modeling limitation.

## 3.2 Physical significance of dielectric loss and wave attenuation

For nonmagnetic tissue under the adopted $e^{+i\omega t}$ convention, the real permittivity stores electric energy whereas conductivity converts field energy into heat. A useful dimensionless comparison is the dielectric loss tangent

$$\tan\delta = \frac{\sigma}{\omega\varepsilon_0\varepsilon_r}, \qquad \omega = 2\pi(2.45\times10^9) \simeq 1.5394\times10^{10}\ \text{rad/s}. \tag{2}$$

For a locally homogeneous, unbounded material, a plane wave has propagation factor $\exp[-(\alpha + i\beta)s]$, where

$$\gamma = \alpha + i\beta = \sqrt{i\omega\mu_0\,(\sigma + i\omega\varepsilon_0\varepsilon_r)}, \quad \delta_E = \frac{1}{\alpha}, \quad \lambda_{\text{eff}} = \frac{2\pi}{\beta}. \tag{3}$$

Table 5 evaluates these *analytical material-level indicators*, not additional finite-element outputs.

Table 5: Derived electromagnetic material indicators at 2.45 GHz from the specified constant $\varepsilon_r$ and $\sigma$; $\delta_E$ is the $1/e$ *field-amplitude* distance of a uniform plane wave.

| Region | $\tan\delta$ | $\alpha$ [m$^{-1}$] | $\beta$ [m$^{-1}$] | $\delta_E$ [mm] | $\lambda_{\text{eff}}$ [mm] |
|---|---|---|---|---|---|
| Healthy liver 106 | 0.2486 | 48.05 | 392.49 | 20.81 | 16.01 |
| Tumor 107 | 0.2562 | 51.55 | 409.01 | 19.40 | 15.36 |

These values explain why tissue is neither an ideal dielectric nor an ideal conductor at the operating frequency. The attenuation and wavelength scales are comparable with tumor dimensions, making antenna placement and interference important. In the real configuration the near field, cylindrical feed, two radiating slots, material interfaces, and tumor shape invalidate a one-dimensional plane-wave intensity law; $\delta_E$ is *not* the simulated ablation radius. The power-density envelope attenuates approximately as $\exp(-2\alpha s)$ only in that simplified homogeneous plane-wave limit.

### 3.3 $H(\mathrm{curl})$ finite elements and weak form

The field is approximated by first-order Nédélec (edge) finite elements on tetrahedra, implemented using `ND_FECollection(1,3)` and `ParFiniteElementSpace`. These degrees of freedom represent tangential electromagnetic continuity more naturally than ordinary scalar nodal interpolation. For complex test field $\mathbf{v}$ the assembled form is, schematically,

$$\begin{aligned} a(\mathbf{E},\mathbf{v}) = &\int_\Omega \mu^{-1}(\nabla\times\mathbf{E})\cdot(\nabla\times\overline{\mathbf{v}})\,dV - \omega^2\int_\Omega \varepsilon\mathbf{E}\cdot\overline{\mathbf{v}}\,dV \\ &+ i\omega\int_\Omega \sigma\mathbf{E}\cdot\overline{\mathbf{v}}\,dV + \int_{\Gamma_{201}\cup\Gamma_{203}\cup\Gamma_{204}} q\,\mathbf{E}_t\cdot\overline{\mathbf{v}}_t\,dS = \ell(\mathbf{v}), \end{aligned} \tag{4}$$

where $q = ik/\mu$ with $k = \omega\sqrt{\mu\tilde{\varepsilon}}$ on an outgoing absorbing boundary. On metal surfaces, $\mathbf{n}\times\mathbf{E} = \mathbf{0}$ is imposed as an essential boundary condition. The port contributes a nonzero incident forcing to $\ell$ in the `-port tem` mode. The implementation uses the `HERMITIAN` complex-operator convention; statements about the sign of $q$ must be read together with the normal and weak-form conventions in the code.

The outer termination is a *first-order local absorbing (Robin/impedance) condition*, not a perfectly matched layer (PML). Its accuracy, especially for obliquely propagating or near-field components, must be checked separately from the algebraic residual of the linear solver.

### 3.4 Coaxial TEM excitation and nominal input power

The input port is the annulus $x = 0$, $a \le r \le b$, $r = \sqrt{y^2+z^2}$, where $a = 0.25\,\mathrm{mm}$ and $b = 0.80\,\mathrm{mm}$. The prescribed incident transverse electric field for peak voltage $V_{\mathrm{inc},0} = 1\,\mathrm{V}$ is

$$\mathbf{E}_{\mathrm{inc}}(r) = \frac{V_{\mathrm{inc},0}}{r\ln(b/a)}\,\hat{\mathbf{r}}, \qquad Z_0 = \frac{1}{2\pi}\sqrt{\frac{\mu_0}{\varepsilon_0\varepsilon_{r,\mathrm{coax}}}}\ln\frac{b}{a}. \tag{5}$$

The TEM Robin excitation uses a boundary forcing proportional to $2i(k_{\mathrm{coax}}/\mu_0)\mathbf{E}_{\mathrm{inc},t}$. It is assembled by `PortTEMImagForcing` and an MFEM boundary integrator. Unlike the earlier `-port fixed` option, it does not prescribe the *total* electric field. For the modeled coax,

$$Z_0 = 48.948{,}423{,}304\,\Omega, \quad P_{\mathrm{inc},0}^{\mathrm{nom}} = \frac{|V_{\mathrm{inc},0}|^2}{2Z_0} = 0.010{,}214{,}833{,}620\,\mathrm{W}. \tag{6}$$

The factor $1/2$ arises because the excitation uses a *peak* phasor, not an RMS voltage. Nominal incident power should not be confused with power delivered *net* through the port after reflection.

The geometry yields $b/a = 3.2$ and $\ln(b/a) = 1.16315$; thus the radial field increases approximately as $1/r$ toward the inner conductor. This explains why an adequate resolution of the small coaxial port annulus is important for a stable incident field and modal power reconstruction. The modeled coaxial wavelength is $\lambda_{\mathrm{coax}} \simeq c/(f\sqrt{2.03}) \simeq 86.0\,\mathrm{mm}$, considerably longer than the port radial dimensions; it does not follow that the finite-length, slotted outer conductor radiates only a TEM mode. At 10 W nominal incident power the *equivalent incident TEM voltage amplitude* is $V_{\mathrm{peak}} = \sqrt{2Z_0P_{\mathrm{inc}}} = 31.288{,}5\,\mathrm{V}$ and the corresponding traveling-wave peak current is approximately $0.639{,}2\,\mathrm{A}$. These are normalized feed-model amplitudes, *not* device-generator measurements.

### 3.5 Modal projection and reflection coefficient

Electric and magnetic TEM amplitudes are reconstructed from the computed port fields, schematically via projected modal voltages $V_E$ and $V_H$. The traveling-wave amplitudes and reflection coefficient are then

$$V_+ = \frac{V_E + V_H}{2}, \qquad V_- = \frac{V_E - V_H}{2}, \qquad S_{11} = \frac{V_-}{V_+}, \qquad P_\pm = \frac{|V_\pm|^2}{2Z_0}. \tag{7}$$

The code also quantifies the non-TEM component of the tangential field. These projected quantities need not equal the direct total-field surface Poynting integral when the discrete solution contains other components, an imperfectly resolved annular surface, or numerical traces not fully captured by the analytic modal basis. Thus $S_{11}$ is reported as a *preliminary simulated TEM-port estimate*, not a measured antenna reflection coefficient.

The fine-port value $|S_{11}| = 0.25405$ implies a projected fundamental-mode reflected-power fraction $|S_{11}|^2 = 0.06454$ (about 6.45%), a return loss $-20\log_{10}|S_{11}| \simeq 11.90\,\mathrm{dB}$, and an illustrative TEM standing-wave ratio $(1 + |S_{11}|)/(1 - |S_{11}|) \simeq 1.681$. The projected traveling-wave interpretation would associate about 0.645 W of reflected power with an idealized 10 W incident reference. This quantity is *not* interchangeable with the exact measured reflected power or with the residual $10 - P_{106} - P_{107}$: the former is a modal projection, the latter additionally includes numerical feed normalization and outward radiation through the model boundaries.

## 4 Mesh generation and numerical software

### 4.1 Mesh-size fields and final discretization

Gmsh with the OpenCASCADE geometry kernel produces a conforming tetrahedral mesh, optimized with the Netgen 3D option. The final script uses a minimum of local mesh-size fields: a broad near-catheter box, a thin inner-conductor refinement, two fine slot boxes, three annular refinement fields at curved interfaces, a distance-threshold field around the tumor, and a short feed-local box. The principal target values are listed in Table 6. The minimum global permitted size was reduced to 0.10 mm to enable the last field; this setting does not force that size throughout the domain.

Table 6: Target mesh scales in the final R35 + fine-port script.

| Region/field | Target size [mm] | Purpose |
|---|---|---|
| Far field | 4.00 | Limit global element count |
| Near catheter | 0.55 | Preserve near-antenna spatial resolution |
| Inner conductor | 0.12 | Resolve central metallic cylinder |
| Slot boxes | 0.15 | Resolve thin annular radiating gaps |
| Outer coax insulator ($r = 0.80$ mm) | 0.24 | Approximate circular dielectric boundary |
| Outer metal ($r = 1.00$ mm) | 0.25 | Resolve PEC geometry |
| Catheter exterior ($r = 1.20$ mm) | 0.30 | Resolve material interface |
| Tumor surface | 1.40 | Reproduce irregular tissue interface |
| Port annulus, $x \approx 0$–$0.8$ mm | 0.10 | Improve TEM feed integration and projection |

Table 7: Counts for the final saved `antena_tumor_R35_port_h010_netgen_ascii.msh`, determined from the mesh and the working solver log.

| Mesh property | Count | Mesh property | Count |
|---|---|---|---|
| Nodes | 52,095 | All tetrahedra | 294,380 |
| Explicit surface triangles | 40,458 | Metal 101 tetrahedra | 68,227 |
| EM-domain tetrahedra | 226,153 | Global complex edge DOF | 295,630 |
| Real block-system unknowns | 591,260 | Port-201 triangles | 495 |
| PEC-202 triangles | 36,365 | ABC-203 triangles | 3,598 |
| Computed inlet-204 triangles | 168 | Tumor-107 tetrahedra | 51,667 |
| Healthy-106 tetrahedra | 57,967 | Tumor–liver shared faces | 10,056 |

Figure 3 shows the tumor and inserted applicator in the final Netgen-optimized mesh. The surrounding healthy-tissue cylinder is hidden in this view to expose the internal geometry; material 106 remains part of the electromagnetic simulation. The visible surface mesh must therefore be interpreted together with the volume-element counts in Table 7.

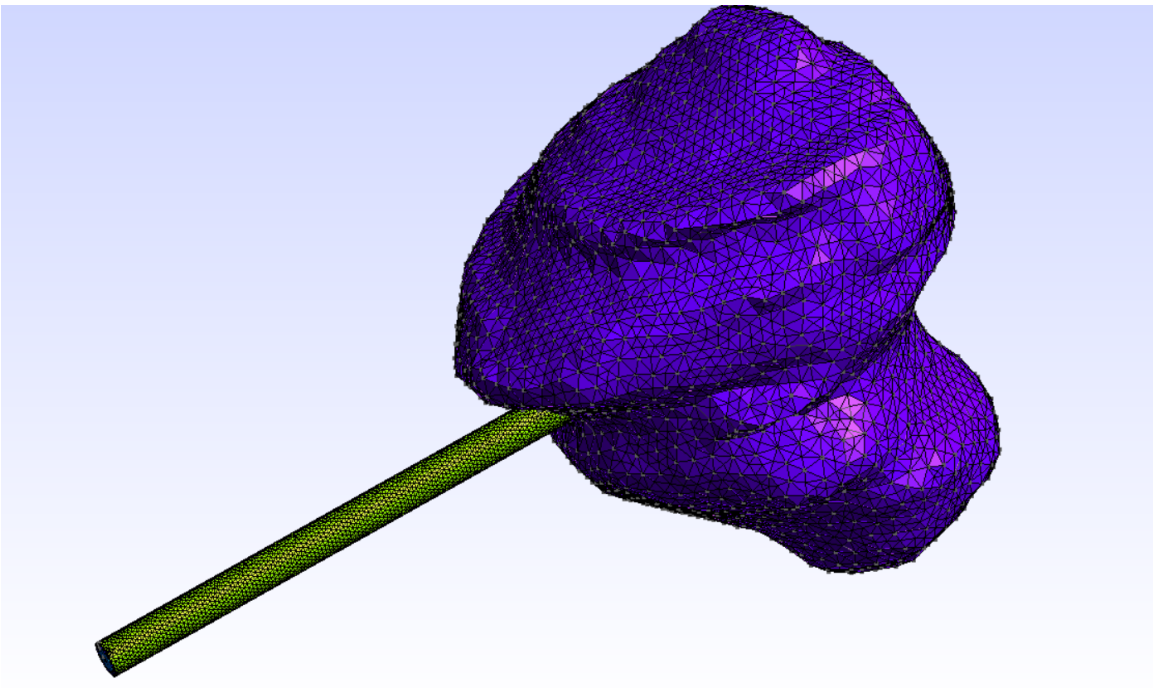


Figure 3: User-supplied view of the final Gmsh tetrahedral mesh after Netgen optimization, used for the R35 fine-port simulation. The tumor and applicator are visible with the surrounding tissue hidden. The complete saved mesh contains 294,380 tetrahedra, including 51,667 in tumor region 107; the electromagnetic submesh contains 226,153 tetrahedra after exclusion of metal.

## 4.2 Topological and element-quality audit

The saved Gmsh 4.1 ASCII mesh was inspected directly rather than relying only on the graphical viewer. All 294,380 tetrahedra have positive signed volume; no practically zero-volume tetrahedron, duplicated coordinate node, duplicated explicit surface triangle, surface triangle without a matching tetrahedral face, or non-manifold tetrahedral face (more than two neighboring cells) was detected. All 495 port triangles neighbor material 102 and all 3,598 exterior ABC triangles neighbor material 106. The catheter inlet contributes 168 boundary triangles with $r = 1.0$–$1.2\,\mathrm{mm}$ in the plane $x = 0$.

A tetrahedral mean-ratio metric (*not* Gmsh's signed inverse condition number, SICN) was computed

as

$$q_{\mathrm{MR}} = \frac{12(3V)^{2/3}}{\sum_{e=1}^{6} \ell_e^2}. \tag{8}$$

For the final mesh, $q_{\min} = 0.223251$, the median is 0.838032, and the first percentile is 0.517503; only three tetrahedra fall below 0.30. An OpenCASCADE shape-orientation warning occurred during Boolean construction, but no resulting topological or inverted-element defect was detected in the generated mesh. This finding does *not* establish that the originating CAD/BREP is defect-free.

### 4.3 Solver and parallel execution

The C++ solver uses MFEM 4.10, MPI-distributed meshes and forms, HYPRE-based parallel matrices, and MUMPS 5.8.2 for the sparse direct solution. A complex operator is mapped to a real $2 \times 2$ block system, with unknown components corresponding to $\Re\mathbf{E}$ and $\Im\mathbf{E}$. Two MPI ranks were used. In development, a serial iterative GMRES attempt reached 600 iterations without satisfying its convergence criterion on an earlier configuration. A direct solver was adopted to avoid interpreting an unconverged iterative field as a physical result; this history is not a comparison of equivalent solvers on the final mesh.

The final R35 fine-port solve has a real block dimension of 591,260 and a true relative residual of $3.1747949814 \times 10^{-12}$ (acceptance threshold $10^{-8}$). The representative MUMPS factorization took approximately 17.84 s, with allocated factorization workspace totaling approximately 5,947 MB across the two processes and 4,883 MB effectively used according to MUMPS diagnostics. These are solver-reported figures, not whole-program peak resident-set measurements. **A small linear residual demonstrates accuracy with respect to the discretized algebraic system, not discretization convergence or experimental validity.**

## 5 Energy deposition, power accounting, and visualization

### 5.1 Time-averaged ohmic heat-source density

For isotropic conductivity $\sigma(\mathbf{x})$ and peak electric-field phasor $\mathbf{E} = \mathbf{E}_R + i\mathbf{E}_I$, the cycle-averaged volumetric power deposition is

$$Q(\mathbf{x}) = \tfrac{1}{2}\sigma(\mathbf{x})(\|\mathbf{E}_R(\mathbf{x})\|^2 + \|\mathbf{E}_I(\mathbf{x})\|^2) \quad [\mathrm{W/m^3}], \qquad P_j = \int_{\Omega_j} Q(\mathbf{x})\, dV. \tag{9}$$

The implemented volumetric integration evaluates the *original* complex Nédélec solution at fourth-order tetrahedral quadrature points via MFEM's physical element transformation and performs MPI global reductions. Material 106 uses $\sigma = 1.95\,\mathrm{S/m}$, and 107 uses $\sigma = 2.18\,\mathrm{S/m}$. The remaining electromagnetic regions are assigned zero conductivity and hence zero ohmic $Q$ in the present model. This approximation does not model metallic Joule heating, dielectric losses in the catheter, or convective catheter cooling.

### 5.2 Poynting flux and independent weak energy check

The magnetic field is recovered consistently with the chosen phasor convention,

$$\mathbf{H} = \frac{i}{\omega\mu}\nabla \times \mathbf{E}, \qquad \langle \mathbf{S} \rangle = \tfrac{1}{2}\,\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*), \qquad P_\Gamma = \int_\Gamma \langle \mathbf{S} \rangle \cdot \mathbf{n}\, dS. \tag{10}$$

Positive $P_\Gamma$ denotes outward flux. The net input $P_{201}^{\mathrm{net}} = -P_{201}$ is therefore compared with the two tissue losses and fluxes through boundaries 203 and 204:

$$\Delta P = P_{201}^{\mathrm{net}} - P_{106} - P_{107} - P_{203} - P_{204} - P_{202}. \tag{11}$$

Eight boundary-flux evaluations (quadrature orders 2, 4, 6 and 8, each with a boundary-transformation and a volume-side implementation) were used to check sensitivity to trace-evaluation details. All gave the same reported port flux for the final first-order, affine mesh. As a second, algebraically independent diagnostic, the discrete complex system obeys

$$\frac{\mathrm{Im}(X^H A X)}{2\omega} = \frac{\mathrm{Im}(X^H B)}{2\omega}, \tag{12}$$

within approximately $6.45 \times 10^{-16}$ W on the last run. On the final mesh the net port flux determined from the weak formulation is 9.521,334,44 mW, versus 9.525,390,63 mW from the total Poynting flux; the difference is 4.056,2 µW. The residual of the *strong, pointwise* first-order Robin condition on boundary 203 remains substantial (relative RMS approximately 0.483), as can occur for a condition enforced weakly. It must not be mistaken for the linear-solver residual.

### 5.3 Nominal 10 W incident-power normalization

The completed reference run uses $V_{\mathrm{inc},0} = 1\,\mathrm{V}$ peak. After electromagnetic solution and verification, the field is scaled to an ideal nominal incident power of 10 W using Eq. (6):

$$\alpha_E = \sqrt{\frac{10\,\mathrm{W}}{P_{\mathrm{inc},0}^{\mathrm{nom}}}} = 31.288471776, \qquad \alpha_Q = \alpha_E^2 = 978.96846607, \tag{13}$$

$$\mathbf{E}_{10\mathrm{W}} = \alpha_E \mathbf{E}_0, \qquad Q_{10\mathrm{W}} = \alpha_Q Q_0. \tag{14}$$

The scaling assumes a linear frequency-domain model with fixed dielectric properties. Importantly, $\alpha_E$ is derived from the *prescribed incident peak voltage* and analytical $Z_0$, not from modal incident power reconstructed from the numerical field. Thus 10 W is a nominal input-setting convention. The model does not assert that 10 W is delivered net to tissue or that the reconstructed modal incident power is exactly 10 W.

### 5.4 Conservative DG0 transfer for subsequent heat modeling

The source routine `MakeThermalSourceDG0` constructs a scalar `L2_FECollection(0,3)` field. For each tetrahedral cell $K$ it computes

$$\overline{Q}_K = \frac{1}{|K|}\int_K Q\,dV, \qquad \sum_{K\subset\Omega_j} |K|\overline{Q}_K = \int_{\Omega_j} Q\,dV. \tag{15}$$

This choice is *conservative in the cell-integrated sense*, unlike assigning $Q$ from one arbitrary sampling point. It is not an exact pointwise representation of $Q$: sub-cell hotspots are averaged within each tetrahedron. The output registers $Q$ for 1 Vpeak and 10 W, and the corresponding real/imaginary electric fields, using `ParaViewDataCollection`. The file is written in parallel to `rezultati/antena_tumor_Pinc10W_Q_mpi/`.

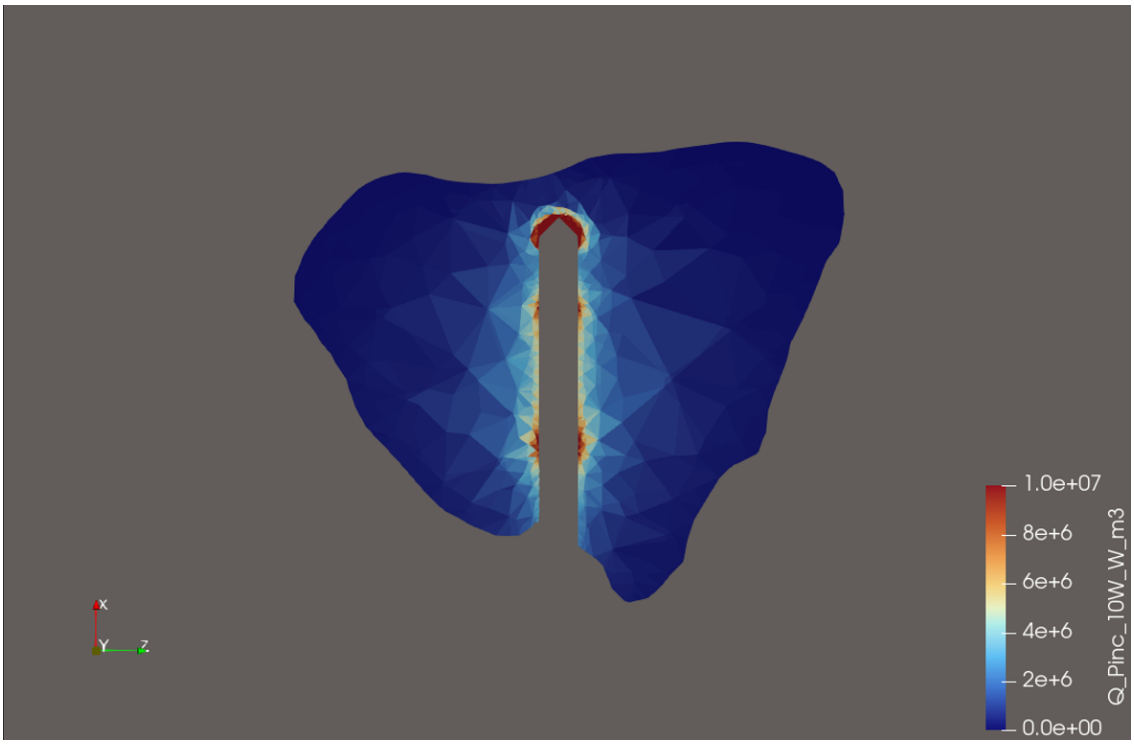



Figure 4: User-supplied ParaView view of the malignant-tissue cross-section and the `Q_Pinc_10W_W_m3` cell-average source near the inserted applicator. The displayed range of 0 to $10^7$ W/m$^3$ was manually chosen for visualization and does not equal the actual maximum of the field. The image depicts *power-density deposition*, not temperature or ablation necrosis.

In Figure 4, the high-$Q$ region is localized near the inserted applicator, while more distant tumor tissue receives substantially less direct electromagnetic heating. The faceted color pattern reflects the piecewise-constant DG0 representation. This is the spatial distribution of the *heat source* in malignant tissue, expressed in W/m$^3$; temperature in °C follows only after solving the time-dependent bioheat problem. The gray background and excluded applicator region do not imply missing malignant-tissue volume.

# 6 Results

## 6.1 Radial-boundary and port-discretization sensitivity

Three completed TEM simulations are available: R27 with a coarse port, R35 with a coarse port, and R35 with a finer port. Table 8 reports direct quantities from their logs. The R27–R35 test explores geometric-boundary sensitivity, while the two R35 meshes explore port-local refinement; Gmsh remeshing is not guaranteed to leave every other element identical.

Table 8: Selected reference-run quantities for a 1 V peak nominal TEM port. Powers are milliwatts unless otherwise stated.

| Quantity | R27, coarse port | R35, coarse port | R35, $h_{\mathrm{port}} = 0.10$ mm |
|---|---|---|---|
| EM tetrahedra | 212,020 | 219,342 | 226,153 |
| Complex ND DOF | 277,047 | 286,349 | 295,630 |
| Port-201 triangles | 120 | 120 | 495 |
| Absorption, tumor 107 [mW] | 5.617171 | 5.644391 | 5.777945 |
| Absorption, liver 106 [mW] | 2.979372 | 3.298415 | 3.420479 |
| Net Poynting input at 201 [mW] | 9.424301 | 9.402269 | 9.525391 |
| Outgoing 203 flux [mW] | 0.717004 | 0.336735 | 0.343411 |
| Outgoing 204 flux [mW] | 0.010806 | 0.009889 | 0.010273 |
| Signed power-balance residual [mW] | +0.099947 | +0.112839 | −0.026717 |
| Absolute relative residual [%] | 1.0605 | 1.2001 | 0.2805 |
| Projected $S_{11}$ [dB] | −11.889 | −11.935 | −11.902 |

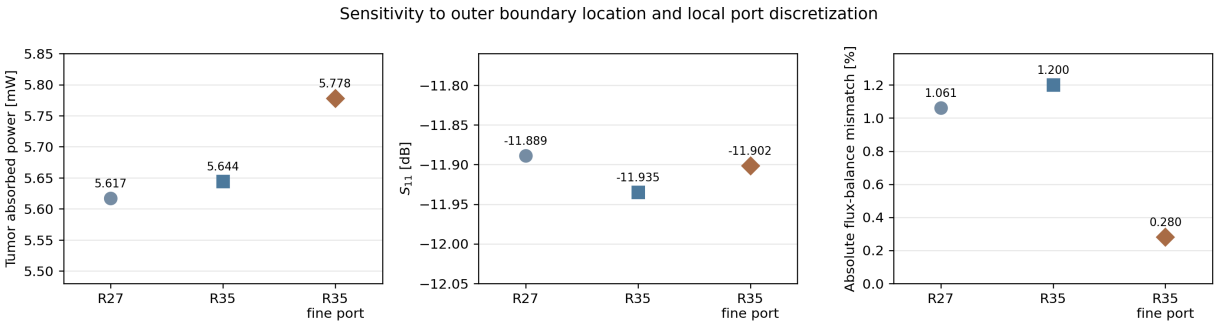



Figure 5: Data reproduced from solver logs: changes in tumor absorption, projected reflection, and surface-flux-balance mismatch across three configurations. This is a *sensitivity comparison*, not a systematic mesh-convergence or mesh-independent error estimate.

Changing the outer tissue radius from 27 to 35 mm changed the tumor-integrated absorbed power by approximately 0.485%, while the projected reflection coefficient varied by about 0.046 dB. The healthy-tissue integral changed more strongly because the *integrated tissue domain itself grew* from approximately 150,107 to 261,834 $\mathrm{mm}^3$: these two full-domain liver integrals must not be interpreted as a comparison over identical physical regions. The enlarged domain reduced outgoing power through boundary 203 from approximately 0.717 to 0.337 mW, consistent with repositioning the artificial surface farther from the applicator, but not by itself proof of reduced reflection error.

The fine-port R35 mesh increases the number of port triangles from 120 to 495, while total EM element count grows by about 3.1%. The absolute power-balance discrepancy decreases from 1.20% to 0.2805%, and the tumor power changes by about 2.37% relative to the coarse R35 mesh. The latter change is large enough that full discretization independence should *not* yet be claimed. The projected $S_{11}$ nevertheless varies by only approximately 0.033 dB between the two R35 meshes.

## 6.2 TEM feed verification

The target geometrical annulus area is $\pi(b^2 - a^2) = 1.814{,}269{,}757{,}4\,\mathrm{mm}^2$. The triangular approximation improves from $1.793{,}615{,}983\,\mathrm{mm}^2$ (120 triangles) to $1.811{,}422{,}620\,\mathrm{mm}^2$ (495 triangles), reducing absolute relative area error from about 1.14% to 0.157%. Table 9 provides a more detailed power and modal comparison.

Table 9: Numerically evaluated TEM-port quantities for the R35 domain at a prescribed 1 Vpeak incident profile.

| Quantity | 120-triangle port | 495-triangle port |
|---|---|---|
| Analytical nominal incident [mW] | 10.214834 | 10.214834 |
| Projected incident [mW] | 9.802280 | 10.075979 |
| Projected reflected [mW] | 0.627816 | 0.650321 |
| Projected net [mW] | 9.174463 | 9.425658 |
| Direct Poynting net [mW] | 9.402269 | 9.525391 |
| Weak-form net [mW] | 9.258939 | 9.521334 |
| $\lvert V_+ - 1\,\mathrm{V}\rvert$ [V] | 0.020852 | 0.006916 |
| TEM $S_{11}$ [dB] | −11.9349 | −11.9016 |
| Non-TEM tangential $E$ ($L^2$ square fraction) [%] | 3.1393 | 0.7808 |
| Non-TEM $\eta H$ ($L^2$ square fraction) [%] | 2.4436 | 0.7129 |

For the 495-triangle port, the projected incident amplitude is $V_+ = 0.9931794 - 0.0011447i$ V; the projected reflected amplitude is $V_- = 0.0290554 + 0.2506396i$ V. Their ratio is approximately $S_{11} = 0.028964 + 0.252394i$, with $|S_{11}| = 0.254051$ and $|S_{11}|^2 = 0.064542$ (approximately 6.45% *projected TEM* reflected fraction). The reconstructed incident power remains about 1.36% below the analytic nominal value. An additional diagnostic requires attention: the finite-element squared norm of the analytic TEM trial profile on the fine port is 5.422604664 versus its analytic value 5.401866425 (relative difference 0.384%); the coarser port happened to match the analytic norm to much higher precision. Thus, not every port metric converges monotonically, and numerical integration and projection formulas merit further scrutiny. The difference between modal net power and total Poynting net flux is about 0.099,733 mW, or 1.05% of the latter. In contrast, the weak-form and direct Poynting net-port powers differ by only 0.004,056 mW (about 0.043%). The different checks measure different aspects of the discretization and should not be collapsed into a single "convergence" statement.

For physical interpretation, the projected $S_{11}$ describes impedance matching of the chosen feed-mode basis; the direct Poynting flux describes the energy crossing the complete numerical port surface; and the weak-form identity checks consistency of the discrete variational operator. Agreement of all three is desirable, but none alone validates wave absorption in a real tissue sample. The 0.2805% surface-flux mismatch is a *power-balance diagnostic* relative to the direct net input (not a bound on electromagnetic field error), and the changing tumor power under fine-port refinement remains a reason to perform further mesh convergence tests.

### 6.3 Deposition at nominal 10 W

The final reference solution has $P_{106} = 3.420{,}479{,}388{,}0\,\mathrm{mW}$ and $P_{107} = 5.777{,}944{,}851{,}8\,\mathrm{mW}$. Applying the nominal amplitude/power factors of Eq. (14) gives Table 11. A greater *fraction of integrated deposited power* occurs within the tumor than the healthy tissue in this chosen computational domain. This fraction does not quantify whether the entire tumor receives an ablative thermal dose.

A useful physical distinction is between *integrated* absorption and *local* absorbed power per mass. With constant density in each material,

$$\mathrm{SAR}(\mathbf{x}) = \frac{Q(\mathbf{x})}{\rho(\mathbf{x})} = \frac{\sigma|\mathbf{E}|^2}{2\rho} \quad [\mathrm{W/kg}], \qquad \overline{\mathrm{SAR}}_j = \frac{P_j}{\rho_j V_j}. \tag{16}$$

Using the actual thermal mesh volumes ($V_{107} = 13.425{,}216{,}4\,\mathrm{cm}^3$ and $V_{106} = 261.833{,}000{,}3\,\mathrm{cm}^3$) and the piecewise densities introduced below gives Table 10. Values are *volume/mass averages*, not local antenna-slot hotspots.

Table 10: Derived absorption metrics at a nominal 10 W incident setting. Tissue volumes/densities correspond to the subsequent thermal domain; these averages do not indicate temperature uniformity.

| Region | Volume [cm$^3$] | Mass [g] | $\langle Q \rangle_V$ [W/m$^3$] | $\overline{\mathrm{SAR}}$ [W/kg] |
|---|---|---|---|---|
| Tumor 107 | 13.4252 | 13.9622 | $4.2133 \times 10^5$ | 405.12 |
| Healthy liver 106 | 261.8330 | 282.5178 | $1.2789 \times 10^4$ | 11.85 |

The tumor receives 62.81% of the integrated loss power while occupying only 4.88% of the modeled *tissue* volume. Consequently, its average specific absorption rate is roughly 34 times the healthy-tissue average. This reflects spatially concentrated deposition rather than selective confinement of a thermal lesion: hotspots can be intense adjacent to the antenna while distant parts of the tumor remain cool. In particular, $P_{106} + P_{107} = 9.005\,\mathrm{W}$ represents 90.05% of the nominal incident setting; the remaining nominal fraction is *not* all reflected power and must be interpreted through the port and surface-flux diagnostics, not as a single physical loss channel.

Table 11: Directly computed and independently post-processed absorbed power. ParaView values are rounded as displayed in the user's screenshots.

| Region | 1 Vpeak reference [W] | 10 W nominal incident [W] | ParaView 3D integral [W] |
|---|---|---|---|
| Tumor (107) | 0.00577794485 | 5.65642581 | 5.65643 |
| Healthy liver (106) | 0.00342047939 | 3.34854146 | 3.34854 |
| Total (106+107) | 0.00919842424 | 9.00496727 | 9.00497 |

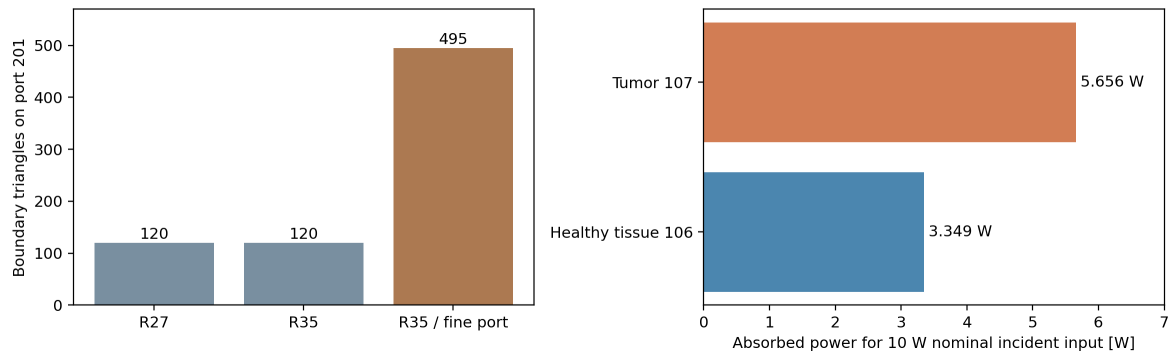
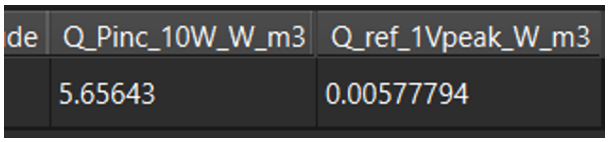


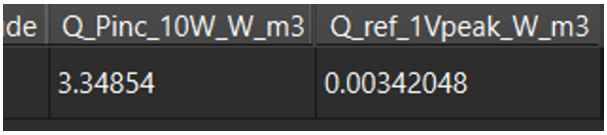

Figure 6: Left: documented refinement of the input TEM annulus. Right: distribution of *integrated* absorbed power between tumor and healthy-tissue regions under the nominal 10 W incident condition. The healthy-tissue cylinder is only a computational surrogate.

The manually inspected ParaView outputs give tumor and healthy-tissue integrals of 5.656,43 W and 3.348,54 W, respectively. Each was generated by thresholding the *entire three-dimensional material volume*, then applying `Integrate Variables`. An earlier attempted integration over a two-dimensional slice yielded units of W/m and was therefore *not* interpreted as total absorbed power; the subsequent volume integrations correct that procedural error.



| de | Q_Pinc_10W_W_m3 | Q_ref_1Vpeak_W_m3 |
|---|---|---|
| | 5.65643 | 0.00577794 |

(a) Tumor 107: 5.656,43 W.



| de | Q_Pinc_10W_W_m3 | Q_ref_1Vpeak_W_m3 |
|---|---|---|
| | 3.34854 | 0.00342048 |

(b) Healthy tissue 106: 3.348,54 W.

Figure 7: Cropped spreadsheet readouts from the user's ParaView checks. The displayed field header retains the original `W_m3` name, but volume integration yields *watts*. The screenshots are documentation of the export audit; Table 11 provides the definitive numerical values.

# 7 Transient thermal model and numerical implementation

## 7.1 Pennes equation, tissue coefficients and physical interpretation

The independently exported electromagnetic heat source $Q_{10\mathrm{W}}(\mathbf{x})$ is loaded without repeating the Maxwell solve. In the tissue-only domain $\Omega_T = \Omega_{106} \cup \Omega_{107}$, Pennes' balance is used as a *one-way coupled* model [14, 4]:

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot (k_T \nabla T) + \rho_b c_b \omega_b (T - T_b) = Q_{10\mathrm{W}}, \quad T(\mathbf{x}, 0) = T_b = 37\,^{\circ}\mathrm{C}. \tag{17}$$

The metabolic term is set to zero. Temperature is represented by first-order continuous nodal $H^1$ elements; $k_T$ and $\rho c_p$ differ between conforming material subregions, while the baseline perfusion rate is the same in regions 106 and 107. The exterior tissue and catheter–tissue boundaries are treated as insulated ($\mathbf{n} \cdot k_T \nabla T = 0$). Perfect thermal insulation at the catheter is a preliminary approximation: no heat storage, conductive exchange with applicator solids or cooling fluid is modeled. The tissue/tumor interface uses continuity of temperature and heat flux. The density and thermal properties in Table 12 are adopted literature *model inputs*, not measurements from this particular lesion [12, 13].

Table 12: Constant material properties used in the completed Pennes simulations. Perfusion rate $\omega_b$ in $\mathrm{s}^{-1}$; blood contributes as a distributed sink, without explicitly meshing blood vessels.

| Domain | $\rho$ [kg/m$^3$] | $c_p$ [J/(kg K)] | $k_T$ [W/(m K)] | $\omega_b$ [s$^{-1}$] |
|---|---|---|---|---|
| Healthy liver 106 | 1079 | 3540 | 0.52 | 0.0036 |
| Tumor 107 | 1040 | 3960 | 0.57 | 0.0036 |
| Blood (sink model) | 1060 | 3639 | — | — |

Writing $u = T - T_b$, the sink coefficient is $g_b = \rho_b c_b \omega_b = 13{,}886.424\,\mathrm{W/(m^3\,K)}$. The thermal diffusivities $a_T = k_T/(\rho c_p)$ are approximately $1.361 \times 10^{-7}$ and $1.384 \times 10^{-7}\,\mathrm{m^2/s}$ for liver and tumor. Hence a characteristic conduction length $\sqrt{a_T t}$ is only around 6.4 mm after 300 s; the factor-of-order-one definition of diffusion length depends on the chosen transient response. Perfusion also introduces a length scale $\sqrt{k_T/g_b}$ of about 6.1 mm (liver) or 6.4 mm (tumor). These estimates *do not determine the actual heated shape*; they explain why diffusion and a distributed blood sink cannot necessarily homogenize temperature throughout an irregular 13.425 cm$^3$ tumor in five minutes.

## 7.2 Finite-element time stepping and energy conservation

At one-second intervals, the solver advances the implicit Euler problem

$$(M_{\rho c} + \Delta t[K_{k_T} + M_{g_b}])\mathbf{u}^{n+1} = M_{\rho c}\mathbf{u}^n + \Delta t\,\mathbf{f}_Q, \quad \Delta t = 1\,\mathrm{s}. \tag{18}$$

The symmetric real system is solved on two MPI ranks using CG with a Jacobi/HYPRE smoother. The initial attempt to set up BoomerAMG failed in the installed HYPRE configuration; the replacement ran successfully. The solver has 109,634 tetrahedra in the tissue-only domain (57,967 liver and 51,667 tumor) and 23,826 global continuous temperature degrees of freedom. Before time integration the conservative DG0 source is verified against the reference $P_{106}$ and $P_{107}$. The CSV-to-mesh mapping is audited by cell positions/volumes, eliminating the need to assume identical element order between VTK and MFEM.

For insulated boundaries and $Q_{\mathrm{met}} = 0$, discrete energy balance requires

$$U^n = \int_{\Omega_T} \rho c_p (T^n - T_b)\, dV, \tag{19}$$

$$\frac{U^{n+1} - U^n}{\Delta t} + P_{\mathrm{perf}}^{n+1} = P_Q, \quad P_{\mathrm{perf}}^{n+1} = \int_{\Omega_T} g_b (T^{n+1} - T_b)\, dV. \tag{20}$$

The reported energy residual rounds to 0.000% in the terminal and is approximately $6.6 \times 10^{-11}$% in the retained sensitivity-summary baseline. This excellent *algebraic/discrete* balance neither establishes time-step convergence nor validates adiabatic boundary physics. The reference input has $P_Q = 9.004{,}967{,}268{,}38$ W; over 300 s the integrated EM source supplies $P_Q t = 2{,}701.49$ J, split between stored tissue heat and the cumulative perfusion sink. At the final instant, with $P_{\mathrm{perf}} \simeq 5.865$ W, the remaining instantaneous storage rate is about 3.140 W. The ratio $5.865/9.005 \simeq 65.1\%$ is a *power* ratio at 300 s, not a time-integrated fraction of emitted heat.

### 7.3 Computed time-dependent temperature and spatial localization

The maximum temperature increases from 37 °C to 85.728 °C in 300 s at nominal 10 W incident power. The calculated maximum crosses 60 °C between 30 and 45 s and reaches 80.068 °C at 180 s. Figure 8 shows the recorded numerical terminal series from the completed 300-step solver; plotted values are terminal-rounded, not an independently interpolated run. Perfusion cooling grows as tissue temperature rises, explaining the declining slope of the maximum-temperature trace.

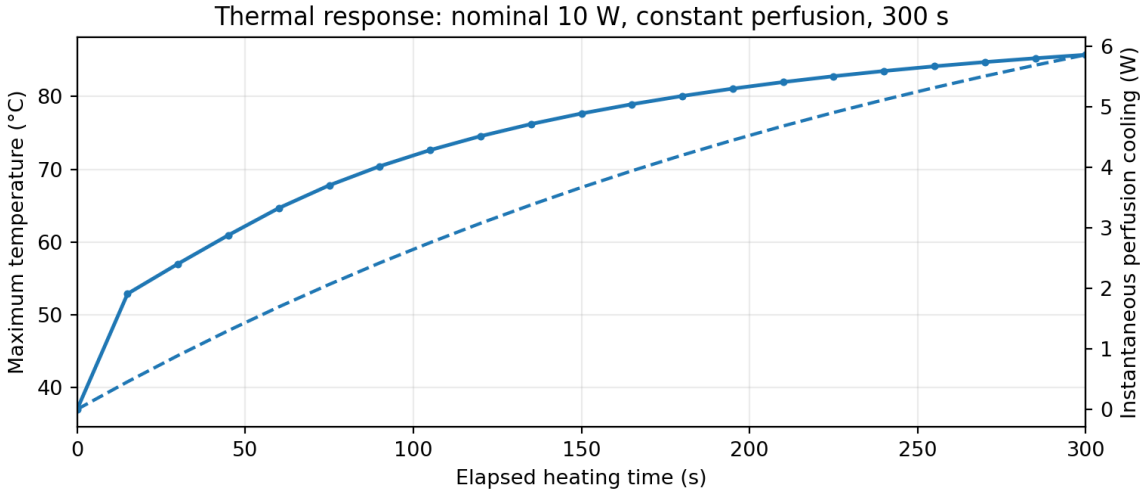



Figure 8: Computed maximum temperature and instantaneous perfusion sink at recorded 15-s intervals for the fixed $Q$ field and constant perfusion. The curves represent the model's time evolution, not patient thermometry.

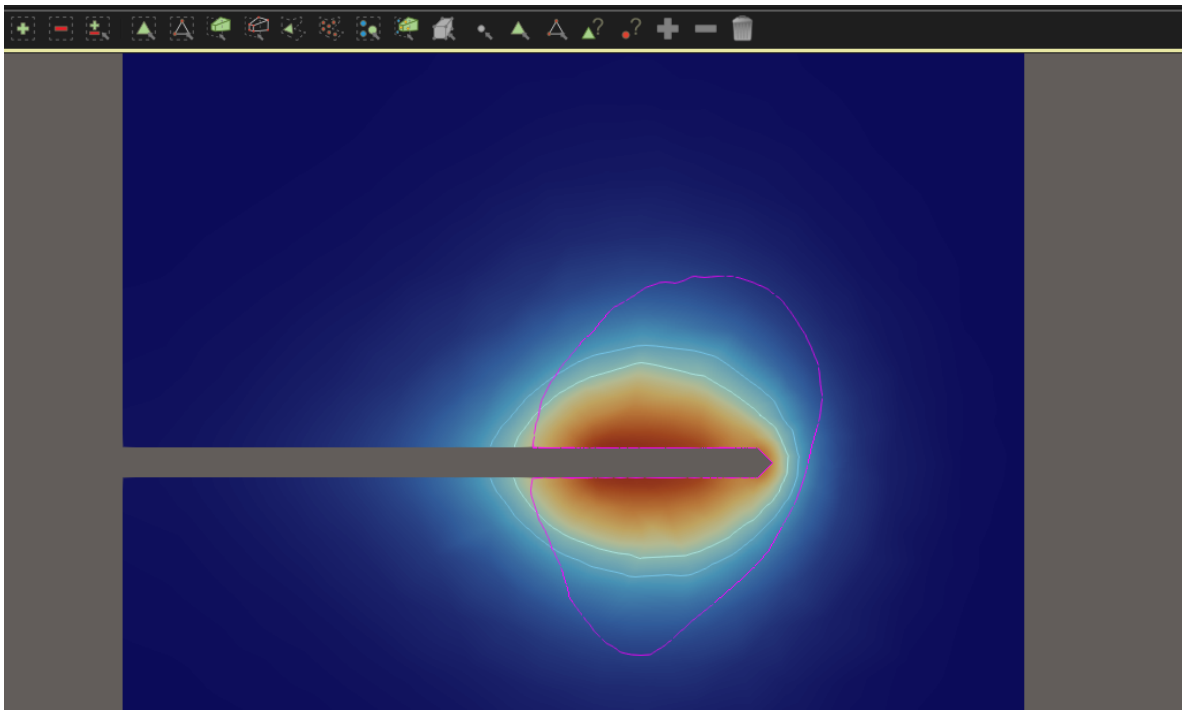


Figure 9: ParaView $z = 0$ temperature cross-section at $t = 300\,\mathrm{s}$, using the user's completed FEM result. The magenta line marks the segmented tumor boundary (material 107); the light-colored curves indicate 55 °C and 60 °C isotherms. The omitted gray applicator/catheter region was excluded from the tissue-only thermal domain. Heat remains localized near the distal applicator: substantial tumor regions lie outside the high-temperature isotherms, whereas some neighboring healthy tissue lies within them. This 2D image is *not* a direct measure of 3D injury volume.

As a separate geometric diagnostic, full-volume ParaView clipping/interpolation of the final temperature field yields $V_{107,T\geq 55^{\circ}C} = 3.874{,}84\,\mathrm{cm}^3$, or approximately 28.86% of the $13.425{,}2\,\mathrm{cm}^3$ tumor. It is not equivalent to the Arrhenius injury coverage because damage depends on the complete temperature history, not solely the temperature at 300 s.

# 8 Arrhenius thermal-injury model

## 8.1 Kinetics, computational method and biological meaning

For each tissue material, a first-order Arrhenius injury measure is accumulated from the transient thermal field:

$$\Omega_j(\mathbf{x},t) = \int_0^t A_j \exp\left[-\frac{E_{a,j}}{R_g\{T_C(\mathbf{x},\tau)+273.15\}}\right] d\tau, \tag{21}$$

$$\theta_j(\mathbf{x},t) = 1-\exp[-\Omega_j(\mathbf{x},t)], \qquad R_g = 8.314462618\ \mathrm{J/(mol\,K)}. \tag{22}$$

We refer to $\theta_j$ as the *model-derived injury fraction*; its literal interpretation as histological necrosis or cell-death probability depends on calibration and the biological assay [9, 10]. The reference classification threshold $\theta_j \geq 0.98$ corresponds to $\Omega_j \geq -\ln(0.02) = 3.912023$, while the optional 99% threshold corresponds to $\Omega_j \geq 4.605170$.

The implemented update uses the trapezoidal time rule at positive second-order tetrahedral integration points,

$$\Omega_q^{n+1} = \Omega_q^n + \frac{\Delta t}{2}\left[k_j(T_q^n)+k_j(T_q^{n+1})\right], \quad k_j(T) = A_j\exp[-E_{a,j}/(R_gT_K)]. \tag{23}$$

Each integration point retains its cumulative $\Omega_q$ across all 300 time steps. The algorithm reports separate quadrature-weighted volumes satisfying the criterion in regions 106 and 107; DG0 fields output a volume average of $\Omega$, an average of $\theta$, and the *fraction of each element's quadrature volume satisfying the threshold*. The averaged $\theta$ of a cell is *not* interchangeable with its thresholded fraction. No feedback from injury to perfusion or EM material properties is modeled.

## 8.2 Two alternative published tumor kinetics

Table 13 records the baseline parameters and alternative liver-cancer cell kinetics. The reference tumor parameters follow in-vitro hepatocellular-carcinoma survival data studied by Reddy *et al.* [9], covering approximately 45 °C to 60 °C; the alternative set is reported by Shah and Bhowmick for *attached HepG2 cells under an isothermal heating protocol*, one of several protocols in the same paper [10]. The liver reference values are used in several computational liver-coagulation models [11, 12]. These pairs have distinct assay/protocol provenance and must *not* be interpreted as two independent estimates of the same measured parameter vector.

Table 13: Arrhenius parameters selected for the parameter comparison; "healthy liver" remains unchanged between the two baseline tumor choices.

| Material / study | $A$ [$s^{-1}$] | $E_a$ [kJ/mol] | Scientific context |
|---|---|---|---|
| Healthy liver 106, reference | $7.390 \times 10^{39}$ | 257.70 | Published tissue injury/coagulation parameters |
| Tumor 107, HCC reference (2013) | $3.247 \times 10^{43}$ | 281.40 | HCC-cell viability; approximate 45 °C to 60 °C |
| Tumor 107, HepG2 attached (2006) | $5.396 \times 10^{36}$ | 248.64 | Attached HepG2 cells, isothermal heating protocol |

The public 3D-IRCADb-01 segmentation identifies a liver lesion but its exact histopathology is not established from the supplied surface mesh. Therefore both tumor kinetic parameterizations are *scenario assumptions*, not tissue constants measured for the same patient. A later patient-specific optimization must first establish a defensible kinetic endpoint and its applicable temperature/time range.

## 8.3 Reference 300-s injury prediction

At the reference input (nominal 10 W, $\omega_b = 0.0036\,s^{-1}$, $\Delta t = 1\,s$), the HCC-parameter model classifies 4.115,013 $cm^3$ of tumor as above the 98% injury criterion. Given $V_{107} = 13.425{,}216\,cm^3$, this is 30.651% coverage; 9.310,203 $cm^3$ does not reach the threshold. In the healthy-liver region, 0.212,284 $cm^3$ is above the same threshold (approximately 0.0811% of the modeled healthy tissue volume). The tumor's volume-averaged continuous $\theta_{107}$ is 0.56397; this *must not be misreported as* 56.4% thresholded tumor coverage. A peak temperature of 85.728 °C is likewise not proof of complete tumor destruction: it occurs locally, while the injury metric is integrated across the entire three-dimensional lesion.

## 8.4 Parameter-sensitivity protocol and quantitative results

Eleven completed computations reuse the same electromagnetic $Q$ distribution and identical thermal time history, changing only the Arrhenius parameters or classification threshold. The supplied reproducibility script checks a constant baseline peak temperature (85.727,703 °C), source power (9.004,967 W), and thermal energy balance across all scenarios. Table 14 transcribes the completed experiment's numerical summary (the complete original machine-readable CSV is in the supplementary package).

Table 14: Completed 300-s Arrhenius sensitivity scan. Coverage refers to tumor volume satisfying $\theta \geq 0.98$ unless noted; the damaged healthy-tissue column is based on the corresponding region-106 kinetic criterion. Perturbation magnitudes are mathematical probes, *not* measured confidence intervals.

| Scenario | Tumor coverage [%] | Healthy injury [cm$^3$] |
|---|---|---|
| HCC reference, 2013 | 30.651 | 0.2123 |
| HepG2 attached, 2006 | 7.481 | 0.2123 |
| Tumor $A \times 0.5$ | 24.514 | 0.2123 |
| Tumor $A \times 2$ | 37.870 | 0.2123 |
| Tumor $E_a - 2\%$ ($A$ fixed) | 58.304 | 0.2123 |
| Tumor $E_a + 2\%$ ($A$ fixed) | 16.050 | 0.2123 |
| Healthy liver $A \times 0.5$ | 30.651 | 0.0947 |
| Healthy liver $A \times 2$ | 30.651 | 0.4721 |
| Tumor $E_a - 10\%$, $k(55°\mathrm{C})$ fixed | 31.552 | 0.2123 |
| Tumor $E_a + 10\%$, $k(55°\mathrm{C})$ fixed | 29.976 | 0.2123 |
| Stricter criterion $\theta \geq 0.99$ | 29.141 | 0.1763 |

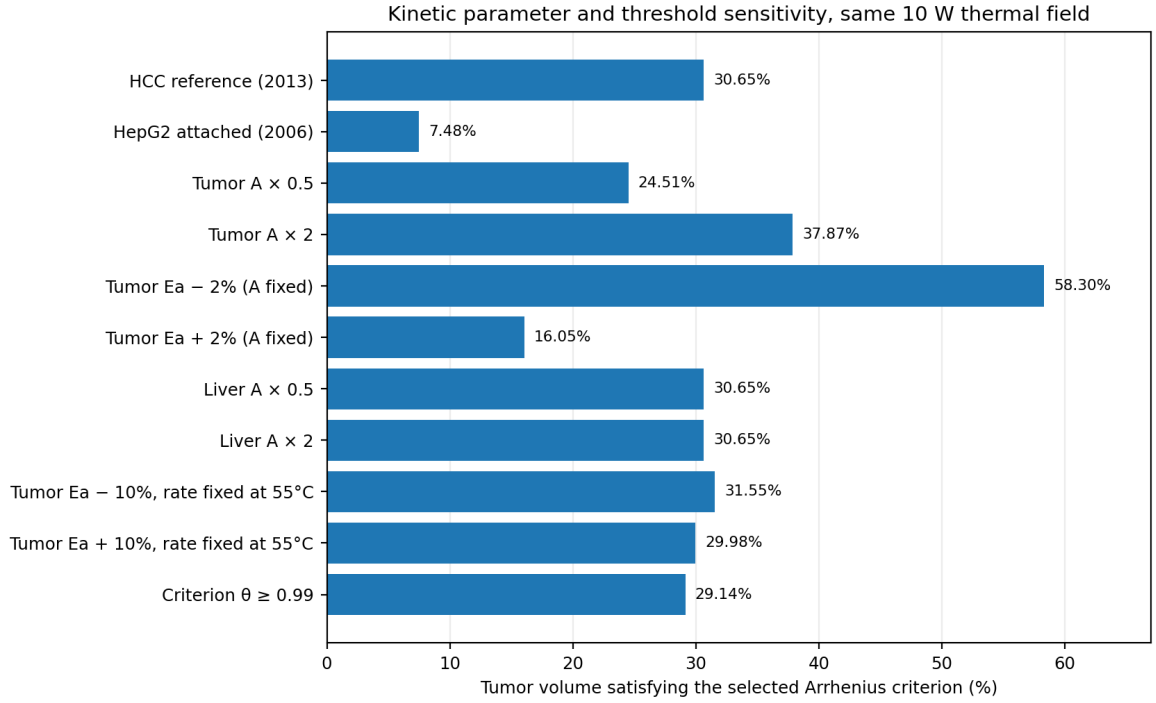



Figure 10: Reproducible sensitivity summary from the user's completed 11-scenario MFEM scan. The normal-tissue kinetic parameters are unchanged in the tumor-only variants; changing the injury criterion affects both tissues. Scenario outcomes are not clinical estimates of uncertainty.

The independent variation of $A_{107}$ by a factor of two changes tumor coverage to 24.514–37.870%; $E_{a,107} \pm 2\%$ at fixed $A_{107}$ causes a much larger range, 16.050–58.304%. The magnitude can be understood directly from logarithmic differentiation:

$$\frac{\partial \ln k}{\partial \ln A} = 1, \qquad \frac{\partial \ln k}{\partial E_a} = -\frac{1}{R_g T_K}. \tag{24}$$

Because $A$ and $E_a$ are jointly inferred from empirical data, independent perturbations can

exaggerate the apparent biological uncertainty. The paired "anchored-55" tests instead use

$$\ln A_{\text{new}} = \ln A_{\text{ref}} + \frac{E_{a,\text{new}} - E_{a,\text{ref}}}{R_g(328.15\,\text{K})} \tag{25}$$

to preserve $k(55\,^\circ\text{C})$ while altering the temperature slope. Although $E_a$ is varied by *ten percent* in either direction, predicted coverage changes much less (29.976–31.552%). This illustrates *dependence on the chosen perturbation family*, not a narrow uncertainty bound. The healthy-liver $A$ perturbations leave tumor coverage unchanged, while healthy tissue injury varies from $0.094{,}7\,\text{cm}^3$ to $0.472{,}1\,\text{cm}^3$; sensitivity of normal-tissue damage matters separately from tumor coverage.

### 8.5 Validity envelope and physiological-temperature extrapolation

A deliberately simple analytic check exposes the danger of extrapolating the Arrhenius formula below its experimental calibration range. If 37 °C is held constant for 300 s without any antenna heating, the current parameter pairs give

$$\theta_{37^\circ\text{C},300s} = 1 - \exp\left[-300A\exp\left(-\frac{E_a}{R_g\,(310.15\,\text{K})}\right)\right]. \tag{26}$$

The results are about 8.44% for the healthy-liver pair, 3.88% for the HCC tumor pair, and 0.216% for the alternative HepG2 pair. These numerical results must *not* be interpreted as actual rapid physiological cell killing. They indicate that application of a high-temperature, in-vitro-fitted single-Arrhenius expression at physiological temperature is outside a safely validated interpretation. Likewise, the reference simulation reaches about 85.7 °C, beyond the 45 °C to 60 °C calibration of the chosen HCC study; predictions at those temperatures represent extrapolation. We therefore retain the published parameterizations as explicit, inspectable assumptions, and avoid relabeling the computed volume as observed "necrosis."

## 9 Exploratory Power–Duration Study at Fixed Antenna Placement

### 9.1 Objective and controlled numerical protocol

The preceding electromagnetic–thermal–injury model showed that a nominal 10 W incident-power setting applied for 300 s produced a peak tissue temperature of 85.73 °C, but only 30.651% of the segmented tumor reached the Arrhenius damage criterion $\theta_d \geq 0.98$. A large local temperature maximum therefore did not imply complete coverage of the irregular lesion. The present study varies heating power and duration while holding the *antenna geometry, slot positions, orientation, and insertion position fixed*. The purpose is to quantify the benefits and penalties of changes in the temporal protocol before remeshing for antenna-placement optimization.

The conforming R35 geometry, 2.45 GHz electromagnetic solution, tissue attributes 106 (healthy liver) and 107 (tumor), and spatially resolved DG0 heat-source field $Q_{10\,\text{W}}(\mathbf{x})$ are unchanged. The source integral for the nominal 10 W reference is

$$P_{\text{abs},10\text{W}} = \int_{\Omega_{106}\cup\Omega_{107}} Q_{10\,\text{W}}(\mathbf{x})\,dV = 9.00496726838\,\text{W}. \tag{27}$$

The Pennes model retains temperature-independent thermophysical properties, $\omega_{106} = \omega_{107} = 0.0036\,\text{s}^{-1}$, initial and blood temperature $T_b = 37\,^\circ\text{C}$, adiabatic thermal boundaries, first-order $H^1$ finite elements, and implicit Euler time stepping with $\Delta t = 1\,\text{s}$. The reference HCC Arrhenius parameters are $A_{107} = 3.247 \times 10^{43}\,\text{s}^{-1}$ and $E_{a,107} = 281400\,\text{J/mol}$. The healthy-liver kinetics and

the threshold $\theta_d = 0.98$, equivalent to $\Omega \geq 3.9120$, are unchanged [9]. All injury volumes in this section are numerical *threshold classifications*, not experimentally or histologically established necrosis.

### 9.2 Scaling the electromagnetic source

Under the assumed linear frequency-domain Maxwell formulation with fixed complex permittivities, the electric-field amplitude varies as $\sqrt{P_{\mathrm{inc}}}$, whereas its time-averaged Joule loss varies as $|\mathbf{E}|^2$. It follows that

$$Q_{P_{\mathrm{inc}}}(\mathbf{x}) = \frac{P_{\mathrm{inc}}}{10\,\mathrm{W}}\, Q_{10\,\mathrm{W}}(\mathbf{x}), \qquad P_{\mathrm{abs}}(P_{\mathrm{inc}}) = \frac{P_{\mathrm{inc}}}{10\,\mathrm{W}} P_{\mathrm{abs,10W}}. \tag{28}$$

Thus the source *intensity* changes while the source *shape* does not. The nominally scaled absorbed powers are 9.00497 W, 9.90546 W, and 10.80596 W at 10, 11, and 12 W, respectively. The nominal incident power must not be confused with the net power delivered to tissue; port mismatch and other electromagnetic energy-balance terms were analyzed separately in the main report. This scaling ceases to be strictly valid if tissue dielectric properties, matching, or geometry change during heating.

The power-duration program is compiled as the independent CMake target `bioheat_power_duration_R35`, without changing the previously archived Arrhenius or electromagnetic solvers. An initial one-second calculation at 12 W verified $Q_{12\mathrm{W}}/Q_{10\mathrm{W}} = 1.2$ and

$$P_{106,12\mathrm{W}} = 4.01824975\,\mathrm{W}, \quad P_{107,12\mathrm{W}} = 6.78771097\,\mathrm{W}, \quad P_{\mathrm{abs},12\mathrm{W}} = 10.80596072\,\mathrm{W}. \tag{29}$$

The calculated peak temperature after 1 s was 41.583 °C, consistent with $37 + 1.2(40.819 - 37) = 41.583\,^{\circ}\mathrm{C}$. This short test established numerical consistency before the completed longer-duration comparisons.

### 9.3 Completed power–duration experiments

Six completed HCC-reference numerical scenarios have been recorded at 10–12 W and 300–600 s. Table 15 compares the maximum *temperature anywhere in the tissue domain*, the tumor volume meeting $\theta_d \geq 0.98$, and the corresponding volume in healthy tissue. The segmented tumor volume is $V_{107} = 13.425216\,\mathrm{cm}^3$.

Table 15: Completed power–duration simulations with a fixed antenna and HCC-reference Arrhenius kinetics. Volumes represent tissue satisfying $\theta_d \geq 0.98$ at the end of treatment, not measured necrosis.

| Nominal power [W] | Duration (s) | $T_{\max}$ [°C] | Tumor coverage [%] | Healthy injury volume [cm$^3$] |
|---|---|---|---|---|
| 10 | 300 | 85.73 | 30.651 | 0.2123 |
| 11 | 300 | 90.60 | 34.774 | 0.3226 |
| 12 | 300 | 95.47 | 39.258 | 0.4630 |
| 10 | 450 | 88.65 | 43.668 | 0.7076 |
| 11 | 450 | 93.81 | 48.892 | 1.0094 |
| 10 | 600 | 89.86 | 52.700 | 1.4483 |

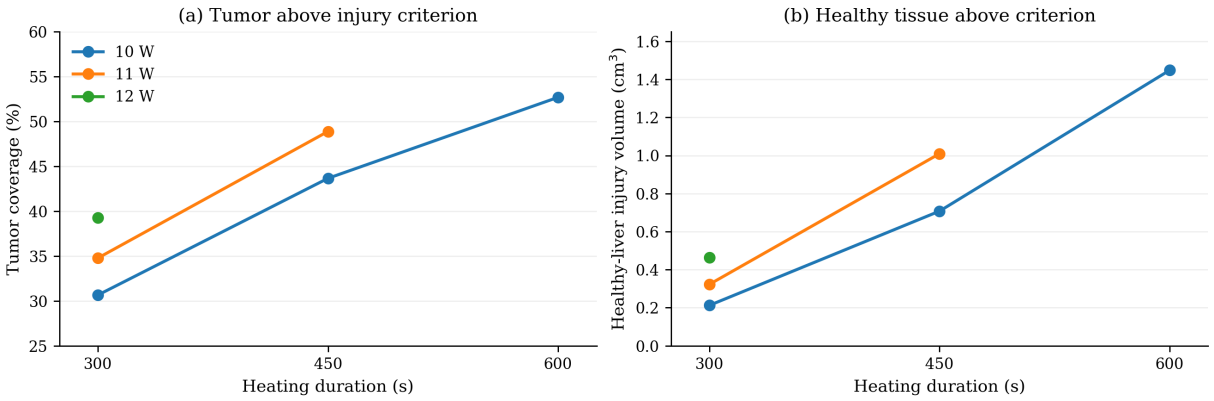



Figure 11: Results from the six completed simulations: (a) fraction of tumor volume above the Arrhenius injury criterion; (b) healthy-liver volume above the same criterion. Lines connect only runs performed at the same nominal power; missing combinations were *not* simulated.

### 9.4 Physical interpretation of duration and power effects

At a fixed 10 W, extending treatment from 300 to 600 s increases tumor coverage from 30.651% to 52.700% (+22.049 percentage points), even though $T_{\max}$ increases by only 4.13 °C. The reason is the distinction between instantaneous temperature and *cumulative* Arrhenius exposure:

$$\Omega(\mathbf{x}, t) = \int_0^t A \exp\left[-\frac{E_a}{R_g T_K(\mathbf{x}, \tau)}\right] d\tau. \tag{30}$$

As tissue warms, heat conduction and perfusion increasingly offset energy input; the local temperature approaches a slowly changing state. However, if the tissue remains hot enough, the time integral of the damage rate continues to rise. The response consequently cannot be assessed from the final peak temperature alone.

The marginal benefit of longer heating decreases in the tested 10 W series: 300–450 s adds 13.017 percentage points of tumor coverage and 0.4953 $\mathrm{cm}^3$ of healthy tissue above threshold; 450–600 s adds only 9.032 points while adding 0.7407 $\mathrm{cm}^3$ of healthy tissue. The modeled incremental tumor coverage therefore diminishes as the associated healthy injury volume grows.

At a fixed 300 s, increasing incident power from 10 to 11 W raises the peak temperature by approximately 4.87 °C, but coverage increases by only 4.123 percentage points; increasing from 11 to 12 W adds approximately another 4.87 °C and 4.484 points. The corresponding healthy-tissue injury volumes increase from 0.2123 to 0.3226 and 0.4630 $\mathrm{cm}^3$. Higher power strengthens the existing deposition hotspot but, under the adopted scaling, does not redirect energy toward more distant portions of the tumor.

The fixed-coefficient Pennes model also predicts a precise temperature scaling when initial and blood temperatures are the same:

$$T_P(\mathbf{x}, t) - T_b = \frac{P}{10\,\mathrm{W}} \left[T_{10\mathrm{W}}(\mathbf{x}, t) - T_b\right]. \tag{31}$$

For example, the 10 W, 300 s result predicts $T_{\max,12\mathrm{W}} \simeq 37 + 1.2(85.728 - 37) = 95.47\,^\circ\mathrm{C}$, which agrees with the recorded 12 W, 300 s result. Likewise, the 11 W, 450 s value of 93.81 °C is consistent with the 10 W, 450 s value of 88.65 °C. This agreement checks the implementation's internal linearity; it is not independent validation of the high-temperature tissue physics.

Two cross-protocol comparisons are instructive. The 10 W, 450 s setting provides 43.668% tumor coverage versus 39.258% for 12 W, 300 s, with a lower peak temperature (88.65 versus 95.47 °C) but greater healthy-tissue injury volume (0.7076 versus 0.4630 $\mathrm{cm}^3$). The 10 W, 600 s setting

attains the highest tumor coverage studied (52.700%), but also the greatest modeled healthy-tissue injury volume (1.4483 $\mathrm{cm}^3$). There is no unambiguous "best" protocol from these two competing measures alone.

### 9.5 Limitations and direction of subsequent optimization

All tested protocols leave a substantial fraction of the tumor below the chosen numerical injury criterion. Even the highest-coverage setting, 10 W for 600 s, leaves 47.300% of the tumor below threshold. Increased total energy delivery cannot compensate indefinitely for a spatially localized electromagnetic source, especially when the tumor extends asymmetrically around the distal antenna region.

The computed maximum temperature reaches or exceeds 90 °C in several scenarios, including 95.47 °C at 12 W for 300 s. At these temperatures, constant dielectric and thermal properties, constant perfusion, neglect of vaporization/dehydration and of heat conduction through the catheter, and one-way coupling limit physical credibility. Likewise, the selected Arrhenius parameters were fitted under specific experimental conditions and have not been calibrated to the histology of this segmented lesion; the prior HCC-versus-HepG2 study showed substantial sensitivity to the kinetic model. These numerical outcomes are therefore *exploratory comparisons*, not clinical predictions or treatment recommendations [9, 14].

Finally, some heated healthy liver could belong to an intended ablative margin: its absolute volume is not by itself a clinical safety measure. Before choosing a treatment protocol, a spatial tumor-margin analysis, numerical mesh/time-step checks, and investigation of perfusion and thermal boundary-condition sensitivity are needed. A useful next step is to map *where* tumor material has not crossed the injury criterion at 300 s. The location of the uncovered volume relative to the two active antenna slots should inform a small set of controlled changes in insertion depth and/or orientation. Each new antenna position requires a fresh Maxwell calculation and conservative transfer of the new heat source, whereas subsequent power and duration studies at that same position can reuse the scaled $Q$ under the linear assumptions. Such work should additionally compare both HCC and HepG2 kinetic parameterizations before drawing any optimization conclusion.

## 10 Discussion

### 10.1 Interpretation of antenna performance

The simulated spatial distribution of $Q$ is concentrated near portions of the applicator and decreases into the surrounding tissue, as expected for a locally driven lossy dielectric system. The exact pattern depends on the slot arrangement, tumor location, material constants and geometric discretization. The selected 1/8/4 mm slot dimensions are supported as a useful dual-slot design concept in [1]; nevertheless, the present simulated $S_{11}$ (about $-11.90$ dB) should not be compared as a controlled performance metric against the experimentally tuned values in that paper. The published study used different geometries, measurements, electromagnetic/thermal assumptions and reference-plane conditions.

The favorable agreement between MFEM's direct $Q$ integrals and ParaView's DG0 volume integrals verifies the *transfer and normalization pipeline*. It does not validate the underlying tissue permittivities or turn power density into temperature. Energy deposition of approximately 9.005 W out of nominal 10 W is physically compatible with the presence of a reflected component and outgoing flux through the exterior boundaries, but one should not infer exact conservation from this rounded comparison alone. The proper reference for numerical energy accounting is the explicitly evaluated net-port/Poynting/weak-form balance.

## 10.2 Verification versus physical validation

The electromagnetic and thermal runs establish several distinct levels of numerical confidence: a conforming, oriented tetrahedral mesh; correct bookkeeping of material and boundary attributes; a small algebraic residual; a discrete weak-form energy identity; improved agreement of strong surface Poynting and weak-form net port flux; a sensitivity test for exterior radius and port resolution; and exact recovery, up to displayed rounding, of integrated losses after DG0 export. However, the following remain *unresolved*:

- No formal $h$-refinement study of the full field or hotspot $Q$ has yet shown convergence of local extrema or of tumor-integrated absorption; localized port refinement changed the tumor integral by approximately 2.37%.
- A first-order absorbing Robin condition, rather than a PML, is used on the truncated cylinder; its residual and boundary-size dependence are not a complete reflection-error bound.
- The modal $S_{11}$ depends on a projected fundamental TEM profile at the feed; higher-order and evanescent content and the reference-plane position have not been experimentally or independently benchmarked.
- The model treats tissues as linear, isotropic and piecewise homogeneous at 2.45 GHz. Patient-specific dielectric properties and temperature dependence have not been evaluated; the thermal extension uses a single constant perfusion value and does not model vessel-specific cooling.
- No phantom measurement, temperature experiment, antenna network-analyzer measurement, independent solver benchmark, thermal-injury histology, or quantified clinical boundary error is available for the present case.
- The lesion is identified as IRCAD case `3Dircadb1.2`, but the complete mask-extraction and patient-to-CAD transformation records remain to be archived; an artificial cylindrical healthy-liver domain is used rather than a complete liver segmentation.

## 10.3 Verification and validation of the completed multiphysics chain

The subsequent transient and damage calculations extend the *numerical* workflow but do not remove its principal physical limitations. The dielectric properties, material geometry, and $Q$ source remain temperature independent, and therefore the coupling is one-way (frequency-domain EM $\rightarrow$ thermal diffusion $\rightarrow$ kinetic damage). In addition, the model omits large perfused vessel geometries, blood-flow advection, catheter conduction, tissue dehydration, vaporization/latent heat, and perfusion shutdown at high temperatures. The maximum calculated temperature is below 100 °C, but omission of all of these mechanisms cannot be justified from that fact alone. The current healthy region is a truncated cylinder rather than a complete liver.

The completed Arrhenius parameter scan demonstrates that the predicted injury volume cannot be identified with a measured coagulation volume. The available HCC and HepG2 parameter pairs were fitted to *different in-vitro viability assays and heating protocols*; the scanned independent parameter changes represent arithmetic sensitivities, not a probabilistic uncertainty distribution. Temporal and spatial convergence studies, anatomical vessel modeling, experimental temperature data, independent thermal validation and histological lesion measurements remain outstanding. Consequently, the present report supports comparisons between explicitly stated *model configurations*, not a recommended clinical treatment.

# 11 Proposed next research stage: antenna-placement optimization

The completed six-protocol power–duration study (Section 9) confirms that changing the incident power or exposure duration alone did not yield full coverage in the present one-position, fixed-coefficient model: tumor coverage ranged from 30.651% to 52.700% with increasing modeled healthy-liver injury. The next step is to map the three-dimensional spatial distribution of tumor material below the selected injury threshold and compare this distribution with the antenna's slot positions and insertion axis. Candidate changes in applicator depth and orientation should then be evaluated individually under matched incident-power and treatment-duration assumptions. Unlike the power scaling used in the completed fixed-geometry sweep, changing placement requires regeneration/validation of the conforming mesh, a new Maxwell solution and a new conservatively transferred $Q$ field. The metrics should remain separate: tumor-criterion coverage, healthy-tissue injury and injury distribution relative to potential ablative margins. Both published tumor kinetic models should be retained as sensitivity cases, with biological-kinetic, thermal-transport and numerical convergence limitations explicitly reported. No clinical treatment prescription follows from these exploratory outputs.

# 12 Conclusions

A three-dimensional, conforming model of a dual-slot coaxial antenna and irregular liver tumor was coupled in sequence to transient thermal diffusion and Arrhenius injury postprocessing. The final electromagnetic mesh contains 226,153 nonmetal tetrahedra and 295,630 complex Nédélec degrees of freedom. The MUMPS algebraic residual is $3.17 \times 10^{-12}$, the direct-flux energy imbalance approximately 0.2805%, and the projected TEM $S_{11}$ approximately $-11.90\,\mathrm{dB}$. A nominal 10 W incident amplitude yields 5.656,43 W of tumor absorption and 3.348,54 W of healthy-tissue absorption; the source was independently checked by the ParaView full-volume integral and a conservative elementwise export.

On the tissue-only conforming thermal mesh, the five-minute Pennes solution gives a final local temperature maximum of 85.728 °C and an instantaneous perfusion sink of approximately 5.865 W; the backward-Euler energy balance is numerically satisfied. The key physical observation is that the high-temperature zone is localized near the distal applicator and does not cover the entire segmented tumor. The reference HCC Arrhenius model predicts that 30.651% of the tumor reaches the chosen 98% model-injury threshold, whereas a different published HepG2 in-vitro model yields 7.481% coverage. Across the completed eleven kinetic and criterion scenarios, neither of these reference settings indicates complete tumor coverage. A further fixed-placement sweep of six completed numerical settings found tumor-criterion coverage ranging from 30.651% at 10 W / 300 s to 52.700% at 10 W / 600 s, accompanied by increasing modeled healthy-tissue injury from $0.212{,}3\,\mathrm{cm}^3$ to $1.448{,}3\,\mathrm{cm}^3$. Raising nominal incident power to 12 W for 300 s yielded 39.258% coverage, $0.463{,}0\,\mathrm{cm}^3$ of healthy-tissue injury and a calculated 95.47 °C maximum. None of these numerical scenarios achieved full injury-criterion tumor coverage; spatial reassessment of antenna placement is therefore motivated, but no clinical prescription can be inferred. Strong kinetic-model dependence, restricted experimental validity ranges, adiabatic tissue boundaries, constant EM/thermal properties, idealized perfusion, absent catheter heat exchange and incomplete convergence/experimental validation must be resolved before estimates of lesion coverage can support clinical conclusions.

## Research ethics, provenance, and reproducibility

The geometry is described as *derived from a publicly available anonymized medical-image resource*; no patient-identifying data are reproduced here. The lesion is identified as case `3Dircadb1.2`. Complete mask-extraction settings, surface-conversion records and the patient-to-CAD transformation should accompany external dissemination. Literature-originated material parameters and any clinical interpretation require author/mentor review. Computations were executed with MFEM 4.10 with OpenMPI and MUMPS 5.8.2 under WSL/Linux, Gmsh with the OpenCASCADE kernel, and ParaView 6.2.0-RC2. Numerical values were transcribed from retained program logs, the completed 11-scenario sensitivity CSV, six-protocol power–duration results and mesh-analysis files; graphics include user-generated mesh and ParaView screenshots, a geometry schematic, and plots of the retained numerical results. Thermal and damage model source, plus numerical sensitivity and power–duration scripts and CSV files, are included in the supplementary directory.

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# A Audit trail: current source functions

The present implementation is substantially more than a single call to a finite-element solver. The following function map documents the scientific purpose of relevant source components. The complete C++ source is included separately in the accompanying archive rather than reproduced line by line.

| Function or class | Numerical/scientific role |
|---|---|
| `PortE`; `PortTEMImagForcing` | Analytic coax TEM electric field and complex Robin excitation profile. |
| `ValidateSerialMesh`, `CheckTEMPortGeometry`, `CheckOpenBoundary` | Mesh/group integrity; correct materials adjacent to port/absorbing boundaries. |
| `ND_FECollection`; `ParSesquilinearForm` | Edge-element complex curl–curl plus dielectric/mass and loss terms. |
| `ComplexHypreParMatrix`; MUMPS solve | Real two-block representation of complex system, distributed direct factorization, true relative residual check. |
| `IntegrateTissuePower` | Fourth-order tetrahedral $Q$ and volume integrals, MPI reductions for regions 106 and 107. |
| `IntegratePoyntingFlux` | Total boundary flux via $\mathbf{H} = i\nabla \times \mathbf{E}/(\omega\mu)$; two transformation methods and multiple quadrature orders. |
| `CalculatePowerBalance`; `CalculateWeakEnergy` | Strong-flux energy accounting and independent algebraic weak identity. |
| `InspectABCFaces` | Per-triangle outgoing-Robin trace residuals and local boundary-flux differences. |
| `WritePortModalDetail` | TEM projection quality, non-modal field remainder, input/reflected voltage and $S_{11}$. |
| `MakeThermalSourceDG0` | Quadrature-based cell-average $Q$ and conservative source transfer. |
| `WriteNominalPower10WCSV`; `SaveField` | Scaled power accounting, CSV output, parallel VTK/ParaView fields. |

# B Thermal and damage implementation audit

The supplied `bioheat_pennes_mpi_arrhenius.cpp` solves on the 106/107 tissue-only submesh with linear continuous $H^1$ elements and an implicit time step. Core functions include `ArrheniusRate`, `AdvanceDamage` and `ReduceDamage`; the source is mapped using a verified position/volume CSV and retained as a piecewise-constant $Q$ field.

At each time step, quadrature-point injury is accumulated using old and new temperatures. The output fields `Damage_Omega_DG0_avg`, `NecroticFraction_DG0_avg` and `DamageThresholdCoverage_DG0` are written to parallel ParaView files. The `analiza_arrhenius_osetljivost.py` orchestration runs each parameterization in a separate working directory and preserves the baseline. The accompanying `arrhenius_sensitivity_summary.csv` is the source of Table 14.

# C Reproducing the electromagnetic solution and heat-source export

## C.1 Source file, dependencies and build target

The electromagnetic source is `supplementary/antena_em_p_TEM_S11_Q_Pinc10W.cpp`. It is C++ source code and must be compiled before execution; the executable used below is named `antena_em_p`. The supplied `supplementary/CMakeLists.txt` explicitly maps that executable name to this source file, and also defines the two thermal targets used below.

Use a Linux or WSL shell with a C++17 compiler, CMake, MPI, and an installed MFEM package configured with MPI, HYPRE and MUMPS support. The recorded calculations used MFEM 4.10 and MUMPS 5.8.2. The MPI implementation linked into MFEM and MUMPS must match the one used to launch the application. `MFEM_DIR` below denotes the directory containing `MFEMConfig.cmake`, not the report directory. The build commands assume an existing configured MFEM installation; they do not install its dependencies. The EM source explicitly rejects builds without MPI or MUMPS support.

Run the following commands from the extracted report directory, replacing the illustrative MFEM path. Retain the resulting `REPORT_ROOT` variable for the subsequent commands in the same shell.

```
# Start in the extracted directory containing the report .tex file.
REPORT_ROOT="$(pwd)"
cmake -S "$REPORT_ROOT/supplementary" \
 -B "$REPORT_ROOT/build-report" \
 -DMFEM_DIR=/absolute/path/to/mfem/lib/cmake/mfem \
 -DCMAKE_CXX_COMPILER=mpicxx -DCMAKE_BUILD_TYPE=Release
cmake --build "$REPORT_ROOT/build-report" \
 --target antena_em_p --parallel 2
```

The mesh used for the reported results is supplied as `supplementary/antena_tumor_R35_port_h010_netgen_ascii.msh`; regenerating it is unnecessary for reproducing that saved discretization. The geometry script imports `tumor_h1_5_107_izvan_katetera_m.brep`, which must remain alongside the `.geo` file if geometry regeneration is required. Remeshing can change the tetrahedral discretization and invalidates reuse of an old cell-source CSV without renewed correspondence checks.

## C.2 Launch the incident-TEM calculation

The program accepts the mesh filename followed by `-port fixed|tem` and an optional `-rtol` value. **Explicitly specify `-port tem`.** The default in the supplied source is the earlier fixed-field

port mode, which does not produce the nominal 10 W TEM heat-source collection discussed in this report. The `-rtol` argument sets the acceptance threshold for the true relative algebraic residual after the direct solve; it is not a mesh-accuracy control. The EM program has no `-power` argument: its 10 W normalization is implemented in the source.

```
# Use a fresh directory: CSV filenames are fixed by the program.
EM_RUN="$REPORT_ROOT/run_em_R35_10W"
mkdir "$EM_RUN"
cd "$EM_RUN"
set -o pipefail
mpirun -np 2 "$REPORT_ROOT/build-report/antena_em_p" \
 "$REPORT_ROOT/supplementary/antena_tumor_R35_port_h010_netgen_ascii.msh" \
 --port tem --rtol 1e-8 2>&1 | tee em_TEM_R35_10W.log
```

If that run directory already exists, select a new name before repeating the calculation. Check that the command succeeds and that the log reports the expected material groups, 226,153 nonmetal tetrahedra, a residual below $10^{-8}$, and successful output. The retained reference residual is $3.17 \times 10^{-12}$; an identical last digit is not a general cross-platform requirement. The factorization requires several GB of memory, in addition to mesh, matrix and other application storage.

The solver first obtains the 1 V peak incident solution and then applies the linear scaling in Eq. (14). For the reference run, expected absorbed powers after normalization are approximately 5.65643 W in tumor 107 and 3.34854 W in healthy tissue 106, totaling 9.00497 W. These values are useful regression checks for this exact mesh and material model, not a substitute for residual and energy-balance checks.

### C.3 Outputs, visualization and the distinction between two CSV files

The current working directory contains diagnostic CSV files, including `fem_port_TEM_S11.csv` for modal port quantities and `fem_izvor_toplote_Pinc10W.csv` for the normalized power summary. The latter is a *key–value summary*, not a table of individual mesh-cell sources, and cannot be passed directly to the thermal solver.

In ParaView, open `antena_tumor_Pinc10W_Q_mpi.pvd` from the directory `rezultati/antena_tumor_Pinc10W_Q_mpi/` and keep its associated parallel VTK files together. Select `Q_Pinc_10W_W_m3` for the normalized heat-source field. The similarly named `Q_ref_1Vpeak_W_m3` belongs to the unscaled reference excitation. Likewise, `E_real` and `E_imag` are reference fields, while `E_real_Pinc_10W` and `E_imag_Pinc_10W` are scaled fields. A slice is appropriate for displaying spatial patterns, but absorbed power must be integrated over the full three-dimensional material region, not over that slice.

### C.4 Export the conservative cell source for bioheat

The provided exporter reads the saved parallel collection and creates the separate cellwise file `Q_Pinc10W_cells.csv`. Run it with ParaView's `pvpython`; alternatively, use Python with NumPy and the required VTK Python modules installed.

```
# Continue from the EM run directory.
PVD="$EM_RUN/rezultati/antena_tumor_Pinc10W_Q_mpi/antena_tumor_Pinc10W_Q_mpi.pvd"
pvpython "$REPORT_ROOT/supplementary/export_q_from_paraview.py" \
 "$PVD" "$EM_RUN/Q_Pinc10W_cells.csv"
```

The columns are `cx_m`, `cy_m`, `cz_m`, `volume_m3`, and `q_W_m3`. For this particular reference mesh the exporter expects 226,153 tetrahedra, including lossless cells with zero $Q$, and checks the integrated power against 9.0049672684 W with a relative tolerance of $10^{-3}$. The downstream thermal solver retains only tissue regions 106/107 (109,634 tetrahedra). The exporter checks cell representation and power conservation and refuses incompatible output; a failed check must be

investigated rather than bypassed. A different mesh or physical model requires a new EM solution and explicitly justified new reference checks.

# D Thermal and damage reproduction commands

Build the thermal target from the supplied Arrhenius-enabled source and pass it the *cellwise* CSV from Appendix C. The following commands continue in the same shell:

```
cmake --build "$REPORT_ROOT/build-report" \
 --target bioheat_pennes_mpi --parallel 2
THERMAL_RUN="$REPORT_ROOT/run_thermal_R35_10W_300s"
mkdir "$THERMAL_RUN"
cd "$THERMAL_RUN"
mpirun -np 2 "$REPORT_ROOT/build-report/bioheat_pennes_mpi" \
 "$REPORT_ROOT/supplementary/antena_tumor_R35_port_h010_netgen_ascii.msh" \
 "$EM_RUN/Q_Pinc10W_cells.csv" --tend 300 --dt 1 \
 --omega106 0.0036 --omega107 0.0036 \
 --damage-threshold 0.98 --save-every 15 \
 2>&1 | tee bioheat_R35_10W_300s.log
```

The thermal solver does not import the `.pvd` file directly. Its source CSV represents a specific mesh revision and must not be reused on another geometry without a fresh coordinate/volume correspondence check. Outputs include `bioheat_R35_10W_arrhenius_energy.csv`, `bioheat_R35_10W_arrhenius_damage.csv` and a parallel VTK collection in `rezultati/bioheat_R35_10W_arrhenius/`. Retain the logs and generated files when regenerating plots.

The archived `analiza_arrhenius_osetljivost.py` driver still uses the original research-directory layout: it searches for the mesh in `../Antena/`, the executable in `build-mumps-app/`, and the cell CSV beside the script. These paths must be adapted to the actual mesh, executable and CSV before running its `-all` option from this archive. Its presence documents the parameter scan; it is not a path-independent launcher for the extracted report.

# E Placement power–duration study reproducibility

The power–duration experiments were carried out in an isolated solver target `bioheat_power_duration_R35`, using the same immutable 10 W DG0 source CSV, with the scaled absorption field $Q_P = (P/10\,\mathrm{W})Q_{10\mathrm{W}}$. The accompanying files `supplementary/bioheat_pennes_mpi_power.cpp`, `supplementary/sweep_power_duration.py` (development script), and `supplementary/results_power_duration_6_scenarios.csv` preserve the underlying implementation and final transcribed study results. An example command for recreating a single case is shown below; actual executable locations depend on the local build and run directories.

```
cmake --build "$REPORT_ROOT/build-report" \
 --target bioheat_power_duration_R35 --parallel 2
mkdir "$REPORT_ROOT/run_power_11W_450s"
cd "$REPORT_ROOT/run_power_11W_450s"
mpirun -np 2 "$REPORT_ROOT/build-report/bioheat_power_duration_R35" \
 "$REPORT_ROOT/supplementary/antena_tumor_R35_port_h010_netgen_ascii.msh" \
 "$EM_RUN/Q_Pinc10W_cells.csv" --power 11 --tend 450 --dt 1 \
 --omega106 0.0036 --omega107 0.0036 \
 --damage-threshold 0.98 --save-every 15
```

For reproducibility, the exact user-executed version of the sweep driver and its complete run logs should be archived with the final release; included scripts may reflect development versions. The six-protocol data are the recorded numerical outputs, not independently rerun measurements.